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United States Patent	12410186
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Lee; Nam-Jin et al.

Compound, organic optoelectronic element, and display device

Abstract

The present application relates to a compound represented by Chemical Formula 1, an organic optoelectronic diode and a display device. ##STR00001## In Chemical Formula 1, each substituent has the same definition as in the specification.

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Appl. No.:	17/426754
Filed (or PCT Filed):	January 30, 2020
PCT No.:	PCT/KR2020/001417
PCT Pub. No.:	WO2020/159249
PCT Pub. Date:	August 06, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20220098217 A1	Mar. 31, 2022

Foreign Application Priority Data

KR	10-2019-0012107	Jan. 30, 2019
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Publication Classification

Int. Cl.: **C07D519/00** (20060101); **C07D498/04** (20060101); **C07D513/04** (20060101); **C07F9/28** (20060101); **H10K85/60** (20230101); H10K50/16 (20230101); H10K50/18 (20230101)

U.S. Cl.:

CPC **C07D519/00** (20130101); **C07D498/04** (20130101); **C07D513/04** (20130101); **C07F9/28** (20130101); **H10K85/622** (20230201); **H10K85/626** (20230201); **H10K85/654** (20230201); **H10K85/6572** (20230201); **H10K85/6574** (20230201); H10K50/16 (20230201); H10K50/18 (20230201)

Field of Classification Search

CPC: C07D (519/00); C07D (498/04); C07D (513/04); C07F (9/28); H10K (85/622); H10K (85/654); H10K (85/6574); H10K (50/18); H10K (50/16)

USPC: 428/1.1

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Background/Summary

TECHNICAL FIELD

(1) This application claims priority to and the benefits of Korean Patent Application No. 10-2019-0012107, filed with the Korean Intellectual Property Office on Jan. 30, 2019, the entire contents of which are incorporated herein by reference.

(2) The present application relates to a compound, an organic optoelectronic diode and a display device.

BACKGROUND ART

(3) An organic optoelectronic diode is a device capable of interconverting electrical energy and light energy.

(4) An organic optoelectronic diode may be divided into two types depending on the operating principle. One is an optoelectronic diode in which excitons formed by light energy are separated into electrons and holes and electrical energy is generated while the electrons and the holes are each transferred to different electrodes, and the other one is a light emitting diode generating light energy from electrical energy by supplying a voltage or a current to electrodes.

(5) Examples of the organic optoelectronic diode may comprise an organic photoelectric diode, an organic light emitting diode, an organic solar cell, an organic photo conductor drum and the like.

(6) Among these, an organic light emitting diode (OLED) has received much attention recently as demands for flat panel display devices have increased. An organic light emitting diode is a device converting electrical energy to light, and performance of an organic light emitting diode is greatly affected by organic materials disposed between electrodes.

(7) An organic light emitting diode has a structure disposing an organic thin film between two electrodes. When a voltage is applied to an organic light emitting diode having such a structure, electrons and holes injected from the two electrodes bind and pair in the organic thin film, and light emits as these annihilate. The organic thin film may be formed in a single layer or a multilayer as necessary.

(8) A material of the organic thin film may have a light emitting function as necessary. For example, as a material of the organic thin film, compounds capable of forming a light emitting layer themselves alone may be used, or compounds capable of performing a role of a host or a dopant of a host-dopant-based light emitting layer may also be used.

(9) In addition thereto, compounds capable of performing roles of hole injection, hole transfer, electron blocking, hole blocking, electron transfer, electron injection and the like may also be used as a material of the organic thin film.

(10) Development of an organic thin film material has been continuously required for enhancing performance, lifetime or efficiency of an organic light emitting diode.

DISCLOSURE

Technical Problem

(11) One embodiment of the present specification is directed to providing a compound capable of

obtaining an organic optoelectronic diode with high efficiency and long lifetime.

(12) Another embodiment of the present specification is directed to providing an organic optoelectronic diode comprising the compound.

(13) Still another embodiment of the present specification is directed to providing a display device comprising the organic optoelectronic diode.

Technical Solution

(14) One embodiment of the present disclosure provides a compound represented by the following Chemical Formula 1.

(15) ##STR00002##

(16) In Chemical Formula 1, any one of X^{sup.4} and X^{sup.5} is —N—, and the other one is —C(L^{sup.2}).sub.n3Ar^{sup.3}—, Ar^{sup.3} is hydrogen; deuterium; a cyano group; —P(=O)RR'; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, X^{sup.1} to X^{sup.3} are each independently —C(L^{sup.4}).sub.n4Ar^{sup.4}—; —C(L^{sup.5}).sub.n5Ar^{sup.5}—; —O—; or —S—, and any one of X^{sup.1} to X^{sup.3} is —O—; or —S—, Ar^{sup.4} and Ar^{sup.5} are each independently hydrogen; deuterium; a cyano group; —P(=O)RR'; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, Ar^{sup.1} and Ar^{sup.2} are each independently hydrogen; deuterium; a cyano group; —P(=O)RR'; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, L^{sup.1} to L^{sup.5} are each independently a single bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group, R and R' are each independently hydrogen; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, n1 to n5 are each independently one of integers of 0 to 4, and at least any one of Ar^{sup.1} to Ar^{sup.5} is a substituent having electron properties; or a substituent having hole properties.

(17) The compound may be represented by the following Chemical Formula 2.

(18) ##STR00003##

(19) In Chemical Formula 2, X^{sup.1} is —O—; or —S—, Ar^{sup.1} to Ar^{sup.5} are each independently hydrogen; deuterium; a cyano group; —P(=O)RR'; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, L^{sup.1} to L^{sup.5} are each independently a single bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group, n1 to n5 are each independently one of integers of 0 to 4, R and R' are each independently hydrogen; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, and at least any one of Ar^{sup.1} to Ar^{sup.5} is a substituent having electron properties; or a substituent having hole properties.

(20) The compound may be represented by the following Chemical Formula 3.

(21) ##STR00004##

(22) In Chemical Formula 3, X^{sup.1} is —O—; or —S—, Ar^{sup.1} to Ar^{sup.5} are each independently hydrogen; deuterium; a cyano group; —P(=O)RR'; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, L^{sup.1} to L^{sup.5} are each independently a single bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group, n1 to n5 are each independently one of integers of 0 to 4, R and R' are each independently hydrogen; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, and at least any one of Ar^{sup.1} to Ar^{sup.5} is a substituent having electron properties; or a

substituent having hole properties.

(23) The compound may be represented by the following Chemical Formula 4.

(24) ##STR00005##

(25) In Chemical Formula 4, X^{sup.1} is —O—; or —S—, Ar^{sup.1} to Ar^{sup.5} are each independently hydrogen; deuterium; a cyano group; —P(=O)RR'; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, L^{sup.1} to L^{sup.5} are each independently a single bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group, n1 to n5 are each independently one of integers of 0 to 4, R and R' are each independently hydrogen; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, and at least any one of Ar^{sup.1} to Ar^{sup.5} is a substituent having electron properties; or a substituent having hole properties.

(26) Another embodiment of the present specification provides an organic optoelectronic diode comprising an anode and a cathode facing each other, and at least one organic layer disposed between the anode and the cathode, wherein the organic layer comprises the compound.

(27) Still another embodiment of the present specification provides a display device comprising the organic optoelectronic diode.

Advantageous Effects

(28) An organic optoelectronic diode with high efficiency and long lifetime can be obtained.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. 1 to FIG. 3 are sectional diagrams each illustrating an organic light emitting diode according to one embodiment of the present specification. **100**: Substrate **200**: Anode **300**: Organic Material Layer **301**: Hole Injection Layer **302**: Hole Transfer Layer **303**: Light Emitting Layer **304**: Hole Blocking Layer **305**: Electron Transfer Layer **306**: Electron Injection Layer **400**: Cathode

MODE FOR DISCLOSURE

(2) Hereinafter, embodiments of the present disclosure will be described in detail. However, these are for illustrative purposes only, and the present disclosure is not limited thereto, and is only defined by the category of claims to describe later.

(3) In the present specification, “substituted or unsubstituted” means being substituted with one or more substituents selected from the group consisting of deuterium; a halogen group; —CN; a C1 to C60 linear or branched alkyl group; a C2 to C60 linear or branched alkenyl group; a C2 to C60 linear or branched alkynyl group; a C3 to C60 monocyclic or polycyclic cycloalkyl group; a C2 to C60 monocyclic or polycyclic heterocycloalkyl group; a C6 to C60 monocyclic or polycyclic aryl group; a C2 to C60 monocyclic or polycyclic heteroaryl group; —SiRR'R''; —P(=O)RR'; a C1 to C20 alkylamine group; a C6 to C60 monocyclic or polycyclic arylamine group; a C2 to C60 monocyclic or polycyclic heteroarylamine group, and a substituted or unsubstituted alkoxy group, or being unsubstituted, or being substituted with a substituent bonding two or more of the substituents, or being unsubstituted, or being substituted with a substituent linking two or more substituents selected from among the above-mentioned substituents, or being unsubstituted. In addition, these may further form a ring with adjacent substituents.

(4) For example, the “substituent linking two or more substituents” may comprise a biphenyl group. In other words, a biphenyl group may be an aryl group, or interpreted as a substituent linking two phenyl groups. The additional substituents may be further substituted. R, R' and R'' are the same as or different from each other, and each independently hydrogen; deuterium; —CN; a substituted or unsubstituted C1 to C60 linear or branched alkyl group; a substituted or

unsubstituted C3 to C60 monocyclic or polycyclic cycloalkyl group; a substituted or unsubstituted C6 to C60 monocyclic or polycyclic aryl group; or a substituted or unsubstituted C2 to C60 monocyclic or polycyclic heteroaryl group.

(5) According to one embodiment of the present application, the “substituted or unsubstituted” means being substituted with one or more substituents selected from the group consisting of deuterium, a halogen group, $-\text{CN}$, $-\text{SiRR}'\text{R}''$, $-\text{P}(=\text{O})\text{RR}'$, a C1 to C20 linear or branched alkyl group, a C6 to C60 monocyclic or polycyclic aryl group and a C2 to C60 monocyclic or polycyclic heteroaryl group, or being unsubstituted, and R, R' and R'' are the same as or different from each other and each independently hydrogen; deuterium; $-\text{CN}$; a C1 to C60 alkyl group unsubstituted or substituted with deuterium, a halogen group, $-\text{CN}$, a C1 to C20 alkyl group, a C6 to C60 aryl group and a C2 to C60 heteroaryl group; a C3 to C60 cycloalkyl group unsubstituted or substituted with deuterium, halogen, $-\text{CN}$, a C1 to C20 alkyl group, a C6 to C60 aryl group and a C2 to C60 heteroaryl group; a C6 to C60 aryl group unsubstituted or substituted with deuterium, halogen, $-\text{CN}$, a C1 to C20 alkyl group, a C6 to C60 aryl group and a C2 to C60 heteroaryl group; or a C2 to C60 heteroaryl group unsubstituted or substituted with deuterium, halogen, $-\text{CN}$, a C1 to C20 alkyl group, a C6 to C60 aryl group and a C2 to C60 heteroaryl group.

(6) The term “substitution” means a hydrogen atom bonding to a carbon atom of a compound being changed to another substituent, and the position of substitution is not limited as long as it is a position at which the hydrogen atom is substituted, that is, a position at which a substituent can substitute, and when two or more substituents substitute, the two or more substituents may be the same as or different from each other.

(7) In the present specification, the halogen may comprise fluorine, chlorine, bromine or iodine.

(8) In the present specification, the alkyl group comprises a C1 to C60 linear or branched, and may be further substituted with other substituents. The number of carbon atoms of the alkyl group may be from 1 to 60, specifically from 1 to 40, specifically from 1 to 20 and more specifically from 1 to 10. Specific examples thereof may comprise a methyl group, an ethyl group, a propyl group, an n-propyl group, an isopropyl group, a butyl group, an n-butyl group, an isobutyl group, a tert-butyl group, a sec-butyl group, a 1-methyl-butyl group, a 1-ethylbutyl group, a pentyl group, an n-pentyl group, an isopentyl group, a neopentyl group, a tert-pentyl group, a hexyl group, an n-hexyl group, a 1-methylpentyl group, a 2-methylpentyl group, a 4-methyl-2-pentyl group, a 3,3-dimethylbutyl group, a 2-ethylbutyl group, a heptyl group, an n-heptyl group, a 1-methylhexyl group, a cyclopentylmethyl group, a cyclohexylmethyl group, an octyl group, an n-octyl group, a tert-octyl group, a 1-methylheptyl group, a 2-ethylhexyl group, a 2-propylpentyl group, an n-nonyl group, a 2,2-dimethylheptyl group, a 1-ethyl-propyl group, a 1,1-dimethyl-propyl group, an isohexyl group, a 2-methylpentyl group, a 4-methylhexyl group, a 5-methylhexyl group and the like, but are not limited thereto.

(9) In the present specification, the alkenyl group comprises a C2 to C60 linear or branched, and may be further substituted with other substituents. The number of carbon atoms of the alkenyl group may be from 2 to 60, specifically from 2 to 40, and more specifically from 2 to 20. Specific examples thereof may comprise a vinyl group, a 1-propenyl group, an isopropenyl group, a 1-butenyl group, a 2-butenyl group, a 3-butenyl group, a 1-pentenyl group, a 2-pentenyl group, a 3-pentenyl group, a 3-methyl-1-butenyl group, a 1,3-butadienyl group, an allyl group, a 1-phenylvinyl-1-yl group, a 2-phenylvinyl-1-yl group, a 2,2-diphenylvinyl-1-yl group, a 2-phenyl-2-(naphthyl-1-yl)vinyl-1-yl group, a 2,2-bis(diphenyl-1-yl)vinyl-1-yl group, a stilbenyl group, a styrenyl group and the like, but are not limited thereto.

(10) In the present specification, the alkynyl group comprises a C2 to C60 linear or branched, and may be further substituted with other substituents. The number of carbon atoms of the alkynyl group may be from 2 to 60, specifically from 2 to 40, and more specifically from 2 to 20.

(11) In the present specification, the cycloalkyl group comprises a C3 to C60 monocyclic or polycyclic, and may be further substituted with other substituents. Herein, the polycyclic means a

group in which the cycloalkyl group is directly linked to or fused with another cyclic group. Herein, the another cyclic group may be a cycloalkyl group, but may also comprise other types of cyclic groups such as a heterocycloalkyl group, an aryl group and a heteroaryl group. The number of carbon atoms of the cycloalkyl group may be from 3 to 60, specifically from 3 to 40, and more specifically from 5 to 20. Specific examples thereof may comprise a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a 3-methylcyclopentyl group, a 2,3-dimethylcyclopentyl group, a cyclohexyl group, a 3-methylcyclohexyl group, a 4-methylcyclohexyl group, a 2,3-dimethylcyclohexyl group, a 3,4,5-trimethylcyclohexyl group, a 4-tert-butylcyclohexyl group, a cycloheptyl group, a cyclooctyl group and the like, but are not limited thereto.

(12) In the present specification, the alkoxy group may comprise a C1 to C10 alkoxy group, and more specifically, a methoxy group, an ethoxy group, a propoxy group, a butoxy group, a pentoxy group and the like.

(13) In the present specification, the silyl group may be represented by —SiRR'R'' , and R, R' and R'' have the same definitions as above. More specifically, a dimethylsilyl group, a diethylsilyl group, a methylethylsilyl group and the like may be included.

(14) In the present specification, the phosphine oxide group may be represented by —P(=O)RR' , and R and R' have the same definitions as above. More specifically, —P(=O)dimethyl , —P(=O)diethyl , —P(=O)methylethyl and the like may be included.

(15) In the present specification, the fluorenyl group means a substituent comprising various substituents at the number 9 position. Specifically, a concept comprising a fluorenyl group in which the number 9 position is substituted with two hydrogens, two alkyl groups, two aryl groups or two heteroaryl groups may be used. More specifically, a 9-di-H-fluorenyl group, a 9-di-methyl-fluorenyl group, a 9-di-phenyl-fluorenyl group or the like may be used.

(16) In the present specification, the heterocycloalkyl group comprises O, S, Se, N or Si as a heteroatom, comprises a C2 to C60 monocyclic or polycyclic, and may be further substituted with other substituents. Herein, the polycyclic means a group in which the heterocycloalkyl group is directly linked to or fused with another cyclic group. Herein, the another cyclic group may be a heterocycloalkyl group, but may also comprise other types of cyclic groups such as a cycloalkyl group, an aryl group and a heteroaryl group. The number of carbon atoms of the heterocycloalkyl group may be from 2 to 60, specifically from 2 to 40, and more specifically from 3 to 20.

(17) In the present specification, the aryl group comprises a C6 to C60 monocyclic or polycyclic, and may be further substituted with other substituents. Herein, the polycyclic means a group in which the aryl group is directly linked to or fused with another cyclic group. Herein, the another cyclic group may be an aryl group, but may also comprise other types of cyclic groups such as a cycloalkyl group, a heterocycloalkyl group and a heteroaryl group. The aryl group comprises a spiro group. The number of carbon atoms of the aryl group may be from 6 to 60, specifically from 6 to 40, specifically from 6 to 30 and more specifically from 6 to 25. Specific examples of the aryl group may comprise a phenyl group, a biphenyl group, a triphenyl group, a naphthyl group, an anthryl group, a chrysenyl group, a phenanthrenyl group, a perylenyl group, a fluoranthenyl group, a triphenylenyl group, a phenalenyl group, a pyrenyl group, a tetracenyl group, a pentacenyl group, a fluorenyl group, an indenyl group, an acenaphthylenyl group, a benzofluorenyl group, a spirobifluorenyl group, a 2,3-dihydro-1H-indenyl group, a fused cyclic group thereof and the like, but are not limited thereto.

(18) In the present specification, the spiro group is a group comprising a spiro structure, and may be from C15 to C60. For example, the Spiro group may comprise a structure in which a 2,3-dihydro-1H-indene group or a cyclohexane group spiro-bonds to a fluorenyl group. Specifically, the Spiro group may comprise any one of the groups of the following structural formulae.

(19) ##STR00006##

(20) In the present specification, the heteroaryl group comprises S, O, Se, N or Si as a heteroatom, comprises a C2 to C60 monocyclic or polycyclic, and may be further substituted with other

substituents. Herein, the polycyclic means a group in which the heteroaryl group is directly linked to or fused with another cyclic group. Herein, the another cyclic group may be a heteroaryl group, but may also comprise other types of cyclic groups such as a cycloalkyl group, a heterocycloalkyl group and an aryl group. The number of carbon atoms of the heteroaryl group may be from 2 to 60, specifically from 2 to 40, specifically from 2 to 30 and more specifically from 3 to 25. Specific examples of the heteroaryl group may comprise a pyridyl group, a pyrrolyl group, a pyrimidyl group, a pyridazinyl group, a furanyl group, a thiophene group, an imidazolyl group, a pyrazolyl group, an oxazolyl group, an isoxazolyl group, a thiazolyl group, an isothiazolyl group, a triazolyl group, a furazanyl group, an oxadiazolyl group, a thiadiazolyl group, a dithiazolyl group, a tetrazolyl group, a pyranlyl group, a thiopyranlyl group, a diazinyl group, an oxazinyl group, a thiazinyl group, a dioxynyl group, a triazinyl group, a tetrazinyl group, a quinolyl group, an isoquinolyl group, a quinazolinyl group, an isoquinazolinyl group, a quinoxalinyl group, a naphthyridyl group, an acridinyl group, a phenanthridinyl group, an imidazopyridinyl group, diazanaphthalenyl group, a triazaindene group, an indolyl group, an indolizinyl group, a benzothiazolyl group, a benzoxazolyl group, a benzimidazolyl group, a benzothiophene group, a benzofuran group, a dibenzothiophene group, a dibenzofuran group, a carbazolyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a phenazinyl group, a dibenzosilole group, spirobi(dibenzosilole), dihydrophenazinyl group, a phenoxazinyl group, a phenanthridyl group, an imidazopyridinyl group, a thienyl group, an indolo[2,3-a]carbazolyl group, an indolo[2,3-b]carbazolyl group, an indolinyl group, a 10,11-dihydro-dibenzo[b,f]azepine group, 9,10-dihydroacridinyl group, a phenanthrazinyl group, a phenothiathiazinyl group, a phthalazinyl group, a naphthyridinyl group, a phenanthrolinyl group, a benzo[c][1,2,5]thiadiazolyl group, 5,10-dihydrodibenzo[b,e][1,4]azasilinyl, a pyrazolo[1,5-c]quinazolinyl group, a pyrido[1,2-b]indazolyl group, a pyrido[1,2-a]imidazo[1,2-e]indolinyl group, a 5,11-dihydroindeno[1,2-b]carbazolyl group and the like, but are not limited thereto.

(21) In the present specification, the amine group may be selected from the group consisting of a monoalkylamine group; a monoarylamine group; a monoheteroarylamine group; —NH.sub.2; a dialkylamine group; a diarylamine group; a diheteroarylamine group; an alkylarylamine group; an alkylheteroarylamine group; and an arylheteroarylamine group, and although not particularly limited thereto, the number of carbon atoms is preferably from 1 to 30. Specific examples of the amine group may comprise a methylamine group, a dimethylamine group, an ethylamine group, a diethylamine group, a phenylamine group, a naphthylamine group, a biphenylamine group, a dibiphenylamine group, an anthracenylamine group, a 9-methyl-anthracenylamine group, a diphenylamine group, a phenylnaphthylamine group, a ditolylamine group, a phenyltolylamine group, a triphenylamine group, a biphenylnaphthylamine group, a phenylbiphenylamine group, a biphenylfluorenylamine group, a phenyltriphenylenylamine group, a biphenyltriphenylenylamine group and the like, but are not limited thereto.

(22) In the present specification, the arylene group means an aryl group having two bonding sites, that is, a divalent group. Descriptions on the aryl group provided above may be applied thereto except for each being a divalent group. In addition, the heteroarylene group means a heteroaryl group having two bonding sites, that is, a divalent group. Descriptions on the heteroaryl group provided above may be applied thereto except for each being a divalent group.

(23) In the present specification, hole properties refer to properties capable of forming holes by donating electrons when applying an electric field, and means properties of, by having conducting properties along the HOMO level, facilitating injection of holes formed in an anode to a light emitting layer, migration of holes formed in a light emitting layer to an anode and migration in the light emitting layer.

(24) Substituents having hole properties comprise a substituted or unsubstituted C6 to C60 aryl group having hole properties, a substituted or unsubstituted C2 to C60 heteroaryl group having hole properties, a substituted or unsubstituted arylamine group, a substituted or unsubstituted

heteroarylamine group, or the like.

(25) More specifically, the substituted or unsubstituted C6 to C60 aryl group having hole properties may be a substituted or unsubstituted phenyl group, a substituted or unsubstituted naphthyl group, a substituted or unsubstituted phenanthrenyl group, a substituted or unsubstituted anthracenyl group, a substituted or unsubstituted fluorenyl group, a substituted or unsubstituted triphenylenyl group, a substituted or unsubstituted spiro-fluorenyl group, a substituted or unsubstituted terphenyl group, a substituted or unsubstituted pyrenyl group, a substituted or unsubstituted perylenyl group, or a combination thereof.

(26) More specifically, the substituted or unsubstituted C2 to C60 heteroaryl group having hole properties is a substituted or unsubstituted carbazolyl group, a substituted or unsubstituted dibenzofuranyl group, a substituted or unsubstituted dibenzothiophenyl group, a substituted or unsubstituted indolecarbazolyl group, or the like.

(27) The aryl group or the heteroaryl group, a substituent bonding to the nitrogen of the substituted or unsubstituted arylamine group and the substituted or unsubstituted heteroarylamine group may be, more specifically, a substituted or unsubstituted phenyl group, a substituted or unsubstituted naphthyl group, a substituted or unsubstituted anthracenyl group, a substituted or unsubstituted phenanthryl group, a substituted or unsubstituted naphthacenyl group, a substituted or unsubstituted pyrenyl group, a substituted or unsubstituted biphenyl group, a substituted or unsubstituted p-terphenyl group, a substituted or unsubstituted m-terphenyl group, a substituted or unsubstituted chrysenyl group, a substituted or unsubstituted triphenylenyl group, a substituted or unsubstituted perylenyl group, a substituted or unsubstituted indenyl group, a substituted or unsubstituted furanyl group, a substituted or unsubstituted thiophenyl group, a substituted or unsubstituted pyrrolyl group, a substituted or unsubstituted pyrazolyl group, a substituted or unsubstituted imidazolyl group, a substituted or unsubstituted triazolyl group, a substituted or unsubstituted oxazolyl group, a substituted or unsubstituted thiazolyl group, a substituted or unsubstituted oxadiazolyl group, a substituted or unsubstituted thiadiazolyl group, a substituted or unsubstituted pyridyl group, a substituted or unsubstituted pyrimidinyl group, a substituted or unsubstituted pyrazinyl group, a substituted or unsubstituted triazinyl group, a substituted or unsubstituted benzofuranyl group, a substituted or unsubstituted benzothiophenyl group, a substituted or unsubstituted benzimidazolyl group, a substituted or unsubstituted indolyl group, a substituted or unsubstituted quinolinyl group, a substituted or unsubstituted isoquinolinyl group, a substituted or unsubstituted quinazolinyl group, a substituted or unsubstituted quinoxalinyl group, a substituted or unsubstituted naphthyridinyl group, a substituted or unsubstituted benzoxazinyl group, a substituted or unsubstituted benzothiazinyl group, a substituted or unsubstituted acridinyl group, a substituted or unsubstituted phenazinyl group, a substituted or unsubstituted phenothiazinyl group, a substituted or unsubstituted phenoxazinyl group, or a combination thereof.

(28) In addition, electron properties refer to properties capable of receiving electrons when applying an electric field, and means properties of, by having conducting properties along the LUMO level, facilitating injection of electrons formed in a cathode to a light emitting layer, migration of electrons formed in a light emitting layer to a cathode and migration in the light emitting layer.

(29) The substituted or unsubstituted C2 to C60 heteroaryl group having electron properties may be a substituted or unsubstituted imidazolyl group, a substituted or unsubstituted tetrazolyl group, a substituted or unsubstituted quinolinylene group, a substituted or unsubstituted isoquinolinylene group, a substituted or unsubstituted pyridinylene group, a substituted or unsubstituted pyrimidinylene group, a substituted or unsubstituted triazinylene group, a substituted or unsubstituted furanyl group, a substituted or unsubstituted benzofuranyl group, a substituted or unsubstituted isofuranyl group, a substituted or unsubstituted benzoisofuranyl group, a substituted or unsubstituted oxazoline group, a substituted or unsubstituted benzoxazoline group, a substituted or unsubstituted oxadiazoline group, a substituted or unsubstituted benzoxadiazoline group, a

substituted or unsubstituted oxatriazolyl group, a substituted or unsubstituted thiophenyl group, a substituted or unsubstituted benzothiophenyl group, a substituted or unsubstituted isothiazoline group, a substituted or unsubstituted benzoisothiazoline group, a substituted or unsubstituted thiazoline group, a substituted or unsubstituted benzothiazoline group, a substituted or unsubstituted pyridazinyl group, a substituted or unsubstituted benzopyridazinyl group, a substituted or unsubstituted pyrazinyl group, a substituted or unsubstituted benzopyrazinyl group, a substituted or unsubstituted phthalazinyl group, a substituted or unsubstituted benzoquinolinyl group, a substituted or unsubstituted quinoxalinyl group, a substituted or unsubstituted quinazolinyl group, a substituted or unsubstituted acridinyl group, a substituted or unsubstituted phenanthrolinyl group, a substituted or unsubstituted phenazinyl group, or a combination thereof.

(30) More specifically, the substituted or unsubstituted C2 to C60 heteroaryl group having electron properties may be any one of the following Chemical Formulae X-1 to X-5.

(31) ##STR00007##

(32) In one embodiment of the present application, L^{sup.n} may be a direct bond (or a single bond); a substituted or unsubstituted arylene group; or a substituted or unsubstituted heteroarylene group.

(33) In another embodiment, L^{sup.n} may be a direct bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group.

(34) In another embodiment, L^{sup.n} may be a direct bond; a substituted or unsubstituted C6 to C40 arylene group; or a substituted or unsubstituted C2 to C40 heteroarylene group.

(35) In L^{sup.n}, n means a number for distinguishing substituents.

(36) Hereinafter, a compound according to one embodiment will be described.

(37) The compound according to one embodiment is represented by the following Chemical Formula 1.

(38) ##STR00008##

(39) In Chemical Formula 1, any one of X^{sup.4} and X^{sup.5} is —N—, and the other one is —C(L^{sup.3}).sub.n3Ar^{sup.3}—, Ar^{sup.3} is hydrogen; deuterium; a cyano group; —P(=O)RR'; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, X^{sup.1} to X^{sup.3} are each independently any one of —C(L^{sup.4}).sub.n4Ar^{sup.4}; —C(L^{sup.5}).sub.n5Ar^{sup.5}—; —O—; or —S—, and any one of X^{sup.1} to X^{sup.3} is —O—; or —S—, Ar^{sup.4} and Ar^{sup.5} are each independently hydrogen; deuterium; a cyano group; —P(=O)RR'; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, Ar^{sup.1} and Ar^{sup.2} are each independently hydrogen; deuterium; a cyano group; —P(=O)RR'; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, L^{sup.1} to L^{sup.5} are each independently a single bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group, R and R' are each independently hydrogen; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, n1 to n5 are each independently one of integers of 0 to 4, at least any one of Ar^{sup.1} to Ar^{sup.5} is a substituent having electron properties; or a substituent having hole properties.

(40) The compound represented by Chemical Formula 1 has a structure in which the substituent having electron properties or hole properties bonds to the fused ring that is a core.

(41) The compound corresponding to Chemical Formula 1 of the present application allows the fused ring corresponding to the core structure to have an expanded HOMO/LUMO electron cloud, and, by adjusting the HOMO/LUMO energy level therethrough, hole/electron injection and hole/electron transfer abilities are further enhanced, and as a result, a driving voltage of a device using the same may be lowered.

(42) In addition, by introducing various substituents to the structure of Chemical Formula 1,

compounds having unique properties of the introduced substituents may be synthesized. For example, by introducing substituents normally used as hole injection layer materials, hole transfer layer materials, light emitting layer materials, electron transfer layer materials and charge generation layer materials used for manufacturing an organic light emitting diode to the core structure, materials satisfying conditions required for each organic material layer may be synthesized.

(43) In addition, by introducing various substituents to the structure of Chemical Formula 1, the energy band gap may be finely controlled, and meanwhile, properties at interfaces between organic materials are enhanced, and material applications may become diverse.

(44) Meanwhile, the compound has a high glass transition temperature (T_g), and thereby has excellent thermal stability. Such an increase in the thermal stability becomes an important factor in providing driving stability to a device.

(45) More specifically, the compound may be represented by any one of the following Chemical Formulae 2 to 4.

(46) ##STR00009##

(47) In Chemical Formulae 2 to 4, $X_{sup.1}$ is $-O-$; or $-S-$, $Ar_{sup.1}$ to $Ar_{sup.5}$ are each independently hydrogen; deuterium; a cyano group; $-P(=O)RR'$; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, $L_{sup.1}$ to $L_{sup.5}$ are each independently a single bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group, n_1 to n_5 are each independently one of integers of 0 to 4, R and R' are each independently hydrogen; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, and at least any one of $Ar_{sup.1}$ to $Ar_{sup.5}$ is a substituent having electron properties; or a substituent having hole properties.

(48) In one embodiment of the present application, $L_{sup.1}$ to $L_{sup.5}$ may be each independently a single bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group.

(49) In another embodiment, $L_{sup.1}$ to $L_{sup.5}$ may be each independently a single bond; a substituted or unsubstituted C6 to C40 arylene group; or a substituted or unsubstituted C2 to C40 heteroarylene group.

(50) In another embodiment, $L_{sup.1}$ to $L_{sup.5}$ may be each independently a single bond; a C6 to C40 arylene group unsubstituted or substituted with a C2 to C40 heteroaryl group; or a C2 to C40 heteroarylene group unsubstituted or substituted with a C6 to C40 aryl group.

(51) In another embodiment, $L_{sup.1}$ to $L_{sup.5}$ may be each independently a single bond; a substituted or unsubstituted phenylene group; a substituted or unsubstituted biphenylene group; a substituted or unsubstituted anthracene group; a substituted or unsubstituted divalent pyrimidine group; a substituted or unsubstituted divalent pyridine group; or a substituted or unsubstituted divalent triazine group.

(52) In another embodiment, $L_{sup.1}$ to $L_{sup.5}$ may be each independently a single bond; a phenylene group unsubstituted or substituted with a carbazole group; a biphenylene group; an anthracene group; a divalent pyrimidine group unsubstituted or substituted with a phenyl group; a divalent pyridine group unsubstituted or substituted with a phenyl group; or a divalent triazine group unsubstituted or substituted with a phenyl group.

(53) In one embodiment of the present application, $Ar_{sup.1}$ to $Ar_{sup.5}$ may be each independently hydrogen; deuterium; a cyano group; $-P(=O)RR'$; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group.

(54) In another embodiment, $Ar_{sup.1}$ to $Ar_{sup.5}$ may be each independently hydrogen; a cyano group; $-P(=O)RR'$; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or

unsubstituted C2 to C60 heteroaryl group.

(55) In another embodiment, Ar.sup.1 to Ar.sup.5 may be each independently hydrogen; a cyano group; —P(=O)RR'; a substituted or unsubstituted C6 to C40 aryl group; or a substituted or unsubstituted C2 to C40 heteroaryl group.

(56) In another embodiment, Ar.sup.1 to Ar.sup.5 may be each independently hydrogen; a cyano group; —P(=O)RR'; a C6 to C40 aryl group unsubstituted or substituted with one or more substituents selected from the group consisting of a C6 to C40 aryl group and a C2 to C40 heteroaryl group; or a C2 to C40 heteroaryl group unsubstituted or substituted with one or more substituents selected from the group consisting of a C6 to C40 aryl group and a C2 to C40 heteroaryl group.

(57) In another embodiment, Ar.sup.1 to Ar.sup.5 may be each independently hydrogen; a cyano group; —P(=O)RR'; a substituted or unsubstituted phenyl group; a substituted or unsubstituted naphthyl group; a substituted or unsubstituted triphenylenyl group; a substituted or unsubstituted anthracenyl group; a substituted or unsubstituted terphenyl group; a substituted or unsubstituted pyridine group; a substituted or unsubstituted triazine group; a substituted or unsubstituted pyrimidine group; a substituted or unsubstituted quinazoline group; a substituted or unsubstituted phenanthroline group; a substituted or unsubstituted benzimidazole group; a substituted or unsubstituted benzoxazole group; a substituted or unsubstituted carbazole group; a substituted or unsubstituted dibenzofuran group; or a substituted or unsubstituted dibenzothiophene group.

(58) In another embodiment, Ar.sup.1 to Ar.sup.5 may be each independently hydrogen; a cyano group; —P(=O)RR'; a phenyl group unsubstituted or substituted with a carbazole group; a naphthyl group; a triphenylenyl group; an anthracenyl group unsubstituted or substituted with a naphthyl group; a terphenyl group; a pyridine group unsubstituted or substituted with a pyridine group; a triazine group unsubstituted or substituted with one or more substituents selected from the group consisting of a phenyl group, a biphenyl group, a triphenylenyl group, a dibenzofuran group and a dibenzothiophene group; a pyrimidine group unsubstituted or substituted with one or more substituents selected from the group consisting of a phenyl group, a biphenyl group, a triphenylenyl group, a dibenzofuran group and a dibenzothiophene group; a quinazoline group unsubstituted or substituted with a phenyl group or a biphenyl group; a phenanthroline group unsubstituted or substituted with a phenyl group; a benzimidazole group unsubstituted or substituted with a phenyl group; a benzoxazole group; a carbazole group; a dibenzofuran group; or a dibenzothiophene group.

(59) In one embodiment of the present application, Ar.sup.1 and Ar.sup.2 may be hydrogen.

(60) In one embodiment of the present application, Chemical Formula 2 may be represented by any one of the following Chemical Formulae 2-1 to 2-3.

(61) ##STR00010##

(62) In Chemical Formulae 2-1 to 2-3, X.sup.1, Ar.sup.3 to Ar.sup.5, L.sup.3 to L.sup.5, n3 to n5 have the same definitions as in Chemical Formula 2, Ar.sup.6 is a cyano group; —P(=O)RR'; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, R and R' are each independently hydrogen; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, L.sup.6 is a single bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group, at least any one of Ar.sup.3 to Ar.sup.6 is a substituent having electron properties; or a substituent having hole properties, and n6 is one of integers of 0 to 4.

(63) In one embodiment of the present application, Chemical Formula 3 may be represented by any one of the following Chemical Formulae 3-1 to 3-3.

(64) ##STR00011##

(65) In Chemical Formulae 3-1 to 3-3, X.sup.1, Ar.sup.3 to Ar.sup.5, L.sup.3 to L.sup.5, n3 to n5

have the same definitions as in Chemical Formula 3, Ar.^{sup.6} is a cyano group; —P(=O)RR'; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, R and R' are each independently hydrogen; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, L.^{sup.6} is a single bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group, at least any one of Ar.^{sup.3} to Ar.^{sup.6} is a substituent having electron properties; or a substituent having hole properties, and n₆ is one of integers of 0 to 4.

(66) In one embodiment of the present application, Chemical Formula 4 may be represented by any one of the following Chemical Formulae 4-1 to 4-3.

(67) ##STR00012##

(68) In Chemical Formulae 4-1 to 4-3, X.^{sup.1}, Ar.^{sup.3} to Ar.^{sup.5}, L.^{sup.3} to L.^{sup.5}, n₃ to n₅ have the same definitions as in Chemical Formula 4, Ar.^{sup.6} is a cyano group; —P(=O)RR'; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, R and R' are each independently hydrogen; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, L.^{sup.6} is a single bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group, at least any one of Ar.^{sup.3} to Ar.^{sup.6} is a substituent having electron properties; or a substituent having hole properties, and n₆ is one of integers of 0 to 4.

(69) In the examples described above, the descriptions on the substituent having electron properties overlap with the descriptions on the substituent provided above, and are not repeated.

(70) In the examples described above, the descriptions on the substituent having hole properties also overlap with the descriptions provided above.

(71) More specifically, the substituent having hole properties may be any one of the following Chemical Formulae 5-1 to 5-4.

(72) ##STR00013##

(73) In Chemical Formula 5-1 to Chemical Formula 5-4, X is —O— or —S—, and R.^{sup.b} to R.^{sup.e} are hydrogen, deuterium, a cyano group or a substituted or unsubstituted C1 to C60 alkyl group.

(74) In Chemical Formulae 5-1 to 5-4, means a bonding position.

(75) Chemical Formulae 5-1 to 5-4 may be selected considering hole and electron properties of the whole compound.

(76) In one embodiment of the present application, the substituent having electron properties; or the substituent having hole properties may be —P(=O)RR'; a substituted or unsubstituted triphenylenyl group; a substituted or unsubstituted anthracenyl group; a substituted or unsubstituted terphenyl group; a substituted or unsubstituted pyridine group; a substituted or unsubstituted triazine group; a substituted or unsubstituted pyrimidine group; a substituted or unsubstituted quinazoline group; a substituted or unsubstituted phenanthroline group; a substituted or unsubstituted benzimidazole group; a substituted or unsubstituted benzoxazole group; a substituted or unsubstituted carbazole group; a substituted or unsubstituted dibenzofuran group; or a substituted or unsubstituted dibenzothiophene group.

(77) In another embodiment, the substituent having electron properties; or the substituent having hole properties may be —P(=O)RR'; a triphenylenyl group; an anthracenyl group unsubstituted or substituted with a naphthyl group; a terphenyl group; a pyridine group unsubstituted or substituted with a pyridine group; a triazine group unsubstituted or substituted with one or more substituents selected from the group consisting of a phenyl group, a biphenyl group, a triphenylenyl group, a dibenzofuran group and a dibenzothiophene group; a pyrimidine group unsubstituted or substituted

with one or more substituents selected from the group consisting of a phenyl group, a biphenyl group, a triphenylenyl group, a dibenzofuran group and a dibenzothiophene group; a quinazoline group unsubstituted or substituted with a phenyl group or a biphenyl group; a phenanthroline group unsubstituted or substituted with a phenyl group; a benzimidazole group unsubstituted or substituted with a phenyl group; a benzoxazole group; a carbazole group; a dibenzofuran group; or a dibenzothiophene group.

(78) In one embodiment of the present application, R and R' may be each independently hydrogen; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group.

(79) In another embodiment, R and R' may be each independently a substituted or unsubstituted C6 to C60 aryl group.

(80) In another embodiment, R and R' may be each independently a substituted or unsubstituted C6 to C40 aryl group.

(81) In another embodiment, R and R' may be each independently a C6 to C40 monocyclic or polycyclic aryl group.

(82) In another embodiment, R and R' may be each independently a C6 to C20 monocyclic or polycyclic aryl group.

(83) In another embodiment, R and R' may be each independently a C6 to C20 polycyclic aryl group.

(84) In another embodiment, R and R' may be a phenyl group.

(85) The substituted or unsubstituted C6 to C60 aryl group bonding to the core of the compound represented by Chemical Formula 1 may be any one of substituents of the following Group II.

(86) ##STR00014##

(87) In Group II, * means a bonding position.

(88) When the aryl group is a biphenyl or a substituted or unsubstituted fluorenyl group, the monoamine having aromatic group and fluorenyl group-type substituents continuous through two or more direct bonds has high hole mobility, and a low driving value is obtained even when used in a hole transfer layer.

(89) The compound of the one example described above may be represented by any one of compounds of the following Group I.

(90) ##STR00015## ##STR00016## ##STR00017## ##STR00018## ##STR00019##
##STR00020## ##STR00021## ##STR00022## ##STR00023## ##STR00024## ##STR00025##
##STR00026## ##STR00027## ##STR00028## ##STR00029## ##STR00030## ##STR00031##
##STR00032## ##STR00033## ##STR00034## ##STR00035## ##STR00036## ##STR00037##
##STR00038## ##STR00039## ##STR00040## ##STR00041## ##STR00042## ##STR00043##
##STR00044## ##STR00045## ##STR00046## ##STR00047## ##STR00048## ##STR00049##
##STR00050## ##STR00051## ##STR00052## ##STR00053## ##STR00054## ##STR00055##
##STR00056## ##STR00057## ##STR00058## ##STR00059## ##STR00060## ##STR00061##
##STR00062## ##STR00063## ##STR00064## ##STR00065## ##STR00066## ##STR00067##
##STR00068## ##STR00069## ##STR00070## ##STR00071## ##STR00072## ##STR00073##
##STR00074## ##STR00075## ##STR00076## ##STR00077## ##STR00078## ##STR00079##
##STR00080##

(91) ##STR00081## ##STR00082## ##STR00083## ##STR00084## ##STR00085##
##STR00086## ##STR00087## ##STR00088## ##STR00089## ##STR00090## ##STR00091##
##STR00092## ##STR00093## ##STR00094## ##STR00095## ##STR00096## ##STR00097##
##STR00098## ##STR00099## ##STR00100## ##STR00101## ##STR00102## ##STR00103##
##STR00104## ##STR00105## ##STR00106## ##STR00107## ##STR00108## ##STR00109##
##STR00110## ##STR00111## ##STR00112## ##STR00113## ##STR00114## ##STR00115##

(92) ##STR00116## ##STR00117## ##STR00118## ##STR00119## ##STR00120##
##STR00121## ##STR00122## ##STR00123## ##STR00124## ##STR00125## ##STR00126##

##STR00127## ##STR00128## ##STR00129## ##STR00130## ##STR00131## ##STR00132##
##STR00133## ##STR00134## ##STR00135## ##STR00136## ##STR00137## ##STR00138##
##STR00139## ##STR00140## ##STR00141## ##STR00142## ##STR00143## ##STR00144##
##STR00145## ##STR00146## ##STR00147## ##STR00148## ##STR00149## ##STR00150##
##STR00151## ##STR00152## ##STR00153## ##STR00154## ##STR00155## ##STR00156##
##STR00157## ##STR00158## ##STR00159## ##STR00160## ##STR00161## ##STR00162##
##STR00163## ##STR00164## ##STR00165## ##STR00166##

(93) ##STR00167## ##STR00168## ##STR00169## ##STR00170## ##STR00171##
##STR00172## ##STR00173## ##STR00174## ##STR00175## ##STR00176## ##STR00177##
##STR00178## ##STR00179## ##STR00180## ##STR00181## ##STR00182## ##STR00183##
##STR00184## ##STR00185## ##STR00186## ##STR00187## ##STR00188## ##STR00189##
##STR00190## ##STR00191## ##STR00192## ##STR00193## ##STR00194## ##STR00195##
##STR00196## ##STR00197##

(94) ##STR00198## ##STR00199## ##STR00200## ##STR00201## ##STR00202##
##STR00203## ##STR00204## ##STR00205## ##STR00206## ##STR00207## ##STR00208##
##STR00209## ##STR00210## ##STR00211## ##STR00212## ##STR00213## ##STR00214##
##STR00215## ##STR00216## ##STR00217## ##STR00218## ##STR00219## ##STR00220##
##STR00221## ##STR00222## ##STR00223## ##STR00224## ##STR00225## ##STR00226##
##STR00227## ##STR00228## ##STR00229## ##STR00230## ##STR00231## ##STR00232##
##STR00233## ##STR00234## ##STR00235## ##STR00236## ##STR00237## ##STR00238##

(95) ##STR00239## ##STR00240## ##STR00241## ##STR00242## ##STR00243##
##STR00244## ##STR00245## ##STR00246## ##STR00247## ##STR00248## ##STR00249##
##STR00250## ##STR00251## ##STR00252## ##STR00253## ##STR00254## ##STR00255##
##STR00256## ##STR00257## ##STR00258## ##STR00259## ##STR00260## ##STR00261##
##STR00262## ##STR00263## ##STR00264## ##STR00265## ##STR00266## ##STR00267##
##STR00268## ##STR00269## ##STR00270## ##STR00271## ##STR00272## ##STR00273##
##STR00274## ##STR00275## ##STR00276## ##STR00277## ##STR00278## ##STR00279##
##STR00280## ##STR00281## ##STR00282## ##STR00283## ##STR00284## ##STR00285##

(96) The compound or the composition described above may be for an organic optoelectronic diode, and the compound for an organic optoelectronic diode or the composition for an organic optoelectronic diode may be formed using a dry film-forming method such as chemical vapor deposition.

(97) Hereinafter, an organic optoelectronic diode using the compound for an organic optoelectronic diode or the composition for an organic optoelectronic diode described above will be described.

(98) The organic optoelectronic diode is not particularly limited as long as it is a device capable of interconverting electrical energy and light energy, and examples thereof may comprise an organic photoelectric diode, an organic light emitting diode, an organic solar cell, an organic photo conductor drum and the like.

(99) Another embodiment of the present application provides an organic light emitting diode comprising a first electrode; a second electrode provided opposite to the first electrode; and one or more organic material layers provided between the first electrode and the second electrode, wherein one or more layers of the organic material layers comprise the heterocyclic compound represented by Chemical Formula 1.

(100) In one embodiment of the present application, the first electrode may be an anode, and the second electrode may be a cathode.

(101) In another embodiment, the first electrode may be a cathode, and the second electrode may be an anode.

(102) Specific details on the heterocyclic compound represented by Chemical Formula 1 are the same as the descriptions provided above.

(103) In one embodiment of the present application, the organic light emitting diode may be a blue

organic light emitting diode, and the heterocyclic compound according to Chemical Formula 1 may be used as a material of the blue organic light emitting diode.

(104) In one embodiment of the present application, the organic light emitting diode may be a green organic light emitting diode, and the heterocyclic compound according to Chemical Formula 1 may be used as a material of the green organic light emitting diode.

(105) In one embodiment of the present application, the organic light emitting diode may be a red organic light emitting diode, and the heterocyclic compound according to Chemical Formula 1 may be used as a material of the red organic light emitting diode.

(106) The organic light emitting diode of the present disclosure may be manufactured using common organic light emitting diode manufacturing methods and materials except that one or more organic material layers are formed using the heterocyclic compound described above.

(107) The heterocyclic compound may be formed into an organic material layer using a solution coating method as well as a vacuum deposition method when manufacturing the organic light emitting diode. Herein, the solution coating method means spin coating, dip coating, inkjet printing, screen printing, a spray method, roll coating and the like, but is not limited thereto.

(108) Herein, another example of the organic light emitting diode, one example of the organic optoelectronic diode, will be described with reference to accompanying drawings.

(109) FIGS. 1 to 3 illustrate a lamination order of electrodes and organic material layers of an organic light emitting diode according to one embodiment of the present application. However, the scope of the present application is not limited to these diagrams, and structures of organic optoelectronic diodes known in the art may also be used in the present application.

(110) FIG. 1 illustrates an organic light emitting diode in which an anode (200), an organic material layer (300) and a cathode (400) are consecutively laminated on a substrate (100). However, the structure is not limited to such a structure, and as illustrated in FIG. 2, an organic light emitting diode in which a cathode, an organic material layer and an anode are consecutively laminated on a substrate may also be obtained.

(111) FIG. 3 illustrates a case of the organic material layer being a multilayer. The organic light emitting diode according to FIG. 3 comprises a hole injection layer (301), a hole transfer layer (302), a light emitting layer (303), a hole blocking layer (304), an electron transfer layer (305) and an electron injection layer (306). However, the scope of the present application is not limited to such a lamination structure, and as necessary, layers other than the light emitting layer may not be included, and other necessary functional layers may be further included.

(112) In the organic light emitting diode, the compound represented by Chemical Formula 1 may be used as a material of an electron transfer layer, a hole transfer layer, a light emitting layer, or the like.

(113) As the anode material, materials having relatively large work function may be used, and transparent conductive oxides, metals, conductive polymers or the like may be used. Specific examples of the anode material comprise metals such as vanadium, chromium, copper, zinc and gold, or alloys thereof; metal oxides such as zinc oxide, indium oxide, indium tin oxide (ITO) and indium zinc oxide (IZO); combinations of metals and oxides such as ZnO:Al or SnO₂:Sb; conductive polymers such as poly(3-methylthiophene), poly[3,4-(ethylene-1,2-dioxy)thiophene] (PEDT), polypyrrole and polyaniline, and the like, but are not limited thereto.

(114) As the cathode material, materials having relatively small work function may be used, and metals, metal oxides, conductive polymers or the like may be used. Specific examples of the cathode material comprise metals such as magnesium, calcium, sodium, potassium, titanium, indium, yttrium, lithium, gadolinium, aluminum, silver, tin and lead, or alloys thereof; multilayer structure materials such as LiF/Al or LiO₂/Al, and the like, but are not limited thereto.

(115) As the hole injection material, known hole injection materials may be used, and for example, phthalocyanine compounds such as copper phthalocyanine disclosed in U.S. Pat. No. 4,356,429, or starburst-type amine derivatives such as tris(4-carbazoyl-9-ylphenyl)amine (TCTA), 4,4',4''-

tri[phenyl(m-tolyl)amino]triphenylamine (m-MTDATA) or 1,3,5-tris[4-(3-methylphenylphenylamino)phenyl]benzene (m-MTDAPB) described in the literature [Advanced Material, 6, p. 677 (1994)], polyaniline/dodecylbenzene sulfonic acid, poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonate), polyaniline/camphor sulfonic acid or polyaniline/poly(4-styrene-sulfonate) that are conductive polymers having solubility, and the like, may be used.

(116) As the hole transfer material, pyrazoline derivatives, arylamine-based derivatives, stilbene derivatives, triphenyldiamine derivatives and the like may be used, and low molecular or high molecular materials may also be used.

(117) As the electron transfer material, metal complexes of oxadiazole derivatives, anthraquinodimethane and derivatives thereof, benzoquinone and derivatives thereof, naphthoquinone and derivatives thereof, anthraquinone and derivatives thereof, tetracyanoanthraquinodimethane and derivatives thereof, fluorenone derivatives, diphenyldicyanoethylene and derivatives thereof, diphenylquinone derivatives, 8-hydroxyquinoline and derivatives thereof, and the like, may be used, and high molecular materials may also be used as well as low molecular materials.

(118) As examples of the electron injection material, LiF is typically used in the art, however, the present application is not limited thereto.

(119) As the light emitting material, red, green or blue light emitting materials may be used, and as necessary, two or more light emitting materials may be mixed and used. Herein, two or more light emitting materials may be used by being deposited as individual sources of supply or by being premixed and deposited as one source of supply. In addition, fluorescent materials may also be used as the light emitting material, however, phosphorescent materials may also be used. As the light emitting material, materials emitting light by bonding electrons and holes injected from an anode and a cathode, respectively, may be used alone, however, materials having a host material and a dopant material involving in light emission together may also be used.

(120) When mixing light emitting material hosts, same series hosts may be mixed, or different series hosts may be mixed. For example, any two or more types of materials among N-type host materials or P-type host materials may be selected, and used as a host material of a light emitting layer.

(121) The organic light emitting diode according to one embodiment of the present application may be a top-emission type, a bottom-emission type or a dual-emission type depending on the materials used.

(122) Hereinafter, the embodiments described above will be described in more detail through examples. However, the following examples are for illustrative purposes only and do not limit the scope of a right.

(123) Starting materials and reaction materials used in examples and synthesis examples are, unless particularly mentioned otherwise, purchased from Sigma-Aldrich, TCI, Tokyo chemical industry or P&H tech, or synthesized using known methods.

Preparation of Compound for Organic Optoelectronic Diode

[Preparation Example 1] Preparation of Intermediate A1

(124) ##STR00286##

1) Preparation of Intermediate A1-2

(125) After dissolving 2,3-dibromothiophene (40 g, 165.33 mmol) and phenylboronic acid (48.38 g, 396.79 mmol) in toluene (500 ml), EtOH (100 ml) and H₂O (100 ml), Pd(PPh₃)₃ (3.82 g, 3.31 mmol) and K₂CO₃ (137.1 g, 991.98 mmol) were introduced thereto, and the result was stirred for 5 hours under reflux. After the reaction was completed, the result was extracted with MC and distilled water, and after drying the organic layer with MgSO₄, the solvent was removed using a rotary evaporator, and then the result was purified by column chromatography using dichloromethane and hexane as a developing solvent to obtain Intermediate

A1-2 (35 g, 90%).

2) Preparation of Intermediate A1-1

(126) After dissolving Intermediate A1-2 (40 g, 169.25 mmol) in DMF (400 ml), NBS (33.14 g, 186.18 mmol) was introduced thereto, and the result was stirred for 4 hours at room temperature. After the reaction was completed, the result was extracted with MC and distilled water, and after drying the organic layer with MgSO_4 , the solvent was removed using a rotary evaporator, and then the result was purified by column chromatography using dichloromethane and hexane as a developing solvent to obtain Intermediate A1-1 (50 g, 93%).

3) Preparation of Intermediate A1

(127) After dissolving Intermediate A1-1 (50 g, 158.61 mmol) and bis(pinacolato)diboron (60.42 g, 237.92 mmol) in 1,4-dioxane (500 ml), $\text{Pd}(\text{dppf})\text{Cl}_2$ (2.32 g, 3.17 mmol) and KOAc (46.7 g, 475.84 mmol) were introduced thereto, and the result was stirred for 8 hours under reflux. After the reaction was completed, the result was extracted with MC and water, and after drying the organic layer with MgSO_4 , the result was silica gel filtered, and precipitated with MC/MeOH. The precipitates were filtered to obtain Intermediate A1 (51 g, 89%).

(128) Intermediate A was synthesized in the same manner as in Preparation Example 1 using S1 of the following Table 1 instead of 2,3-dibromothiophene, and using Intermediate S2 instead of phenylboronic acid.

(129) TABLE-US-00001 TABLE 1 S1 S2 Intermediate A Yield

S1	S2	Intermediate A	Yield
0	0	0	79%
1	1	1	82%
2	2	2	88%
3	3	3	78%
4	4	4	00%
5	5	5	78%
6	6	6	02%
7	7	7	03%
8	8	8	04%
9	9	9	85%
10	10	10	05%
11	11	11	06%
12	12	12	07%
13	13	13	87%

[Preparation Example 2] Preparation of Intermediate B1

(130) ##STR00308##

1) Preparation of Intermediate B1-2

(131) After dissolving 2,3-diphenylthiophene (30 g, 126.94 mmol) in chloroform (300 ml), the result was cooled to 0° C., bromine (42.6 g, 266.58 mmol) was slowly introduced thereto, and the result was stirred for 3 hours. After the reaction was completed, the result was extracted with MC and distilled water, and after drying the organic layer with MgSO_4 , the solvent was removed using a rotary evaporator, and then the result was purified by column chromatography using dichloromethane and hexane as a developing solvent to obtain Intermediate B1-2 (43 g, 86%).

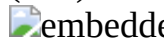
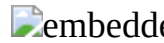
2) Preparation of Intermediate B1-1

(132) After dissolving Intermediate B1-2 (43 g, 109.1 mmol) in anhydrous diethyl ether (400 ml), the result was cooled to -78° C., n-BuLi (45.8 ml, 114.56 mmol) was slowly introduced thereto, and the result was stirred for 1 hour. After the reaction was completed, the result was extracted with EA and a dilute aqueous NH_4Cl solution, and after drying the organic layer with MgSO_4 , the solvent was removed using a rotary evaporator, and then the result was purified by column chromatography using dichloromethane and hexane as a developing solvent to obtain Intermediate B1-1 (27 g, 78%).

3) Preparation of Intermediate B1

(133) After dissolving Intermediate B1-1 (27 g, 85.65 mmol) and bis(pinacolato)diboron (32.63 g, 128.48 mmol) in 1,4-dioxane (300 ml), $\text{Pd}(\text{dppf})\text{Cl}_2$ (2.51 g, 3.43 mmol) and KOAc (25.22 g, 256.96 mmol) were introduced thereto, and the result was stirred for 5 hours under reflux. After the reaction was completed, the result was extracted with MC and water, and after drying the organic layer with MgSO_4 , the result was silica gel filtered, and precipitated with MC/MeOH. The precipitates were filtered to obtain Intermediate B1 (26 g, 83%).

(134) Intermediate B was synthesized in the same manner as in Preparation Example 2 using S3 of the following Table 2 instead of 2,3-diphenylthiophene.

(135) TABLE-US-00002 TABLE 2 S3 Intermediate B Yield 09
 76%   72% 
 84%   78% 
 81%  0  80% 
 77%

[Preparation Example 3] Preparation of Compound 1

(136) ##STR00323## ##STR00324##

1) Preparation of Compound 1-4

(137) After dissolving 5-bromo-2,3-diphenylthiophene (50 g, 158.61 mmol) and 2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)aniline (38.23 g, 174.48 mmol) in dioxane (500 ml) and H.sub.2O (100 ml), Pd(PPh.sub.3).sub.4 (9.16 g, 7.93 mmol) and NaHCO.sub.3 (39.98 g, 475.84 mmol) were introduced thereto, and the result was stirred for 14 hours under reflux. After the reaction was completed, the reaction solution was dissolved by introducing MC thereto, and then extracted with distilled water. After drying the organic layer with MgSO.sub.4, the solvent was removed using a rotary evaporator, and then the result was purified by column chromatography using dichloromethane and hexane as a developing solvent to obtain Compound 1-4 (41 g, 79%).

2) Preparation of Compound 1-3

(138) After dissolving Compound 1-4 (41 g, 125.21 mmol) in MC (500 ml), TEA (38.01 g, 375.64 mmol) was introduced thereto. The temperature was lowered from room temperature to 0° C., 4-bromobenzoyl chloride (30.23 g, 137.74 mmol) dissolved in MC was slowly added dropwise thereto. After the reaction was completed, the result was extracted with MC and distilled water, and after drying the organic layer with MgSO.sub.4, the solvent was removed using a rotary evaporator, and then the result was purified by column chromatography using dichloromethane and hexane as a developing solvent to obtain Compound 1-3 (55 g, 86%).

3) Preparation of Compound 1-2

(139) After dissolving Compound 1-3 (55 g, 107.75 mmol) in nitrobenzene (500 ml), POCl.sub.3 (18.17 g, 118.53 mmol) was slowly added dropwise thereto, and the result was stirred for 4 hours at 150° C. After the reaction was completed, the result was neutralized with an aqueous NaHCO.sub.3 solution and then extracted with MC and distilled water. After drying the organic layer with MgSO.sub.4, the solvent was removed using a rotary evaporator, and then the result was purified by column chromatography using dichloromethane and hexane as a developing solvent to obtain Compound 1-2 (48 g, 90%).

4) Preparation of Compound 1-1

(140) After dissolving Compound 1-2 (48 g, 97.48 mmol) and bis(pinacolato)diboron (37.13 g, 146.21 mmol) in 1,4-dioxane (500 ml), Pd(dppf)Cl.sub.2 (2.85 g, 3.9 mmol) and KOAc (28.7 g, 292.43 mmol) were introduced thereto, and the result was stirred for 5 hours under reflux. After the reaction was completed, the result was extracted with MC and water, and after drying the organic layer with MgSO.sub.4, the result was silica gel filtered, and precipitated with MC/MeOH. The precipitates were filtered to obtain Compound 1-1 (46 g, 87%).

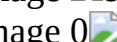
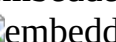
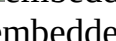
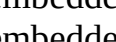
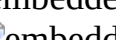
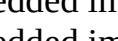
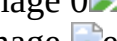
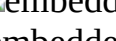
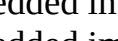
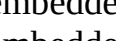
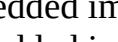
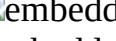
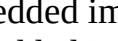
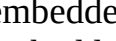
5) Preparation of Compound 1

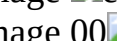
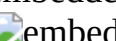
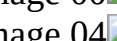
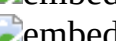
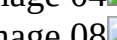
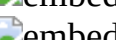
(141) After dissolving Compound 1-1 (10 g, 18.54 mmol) and 4-(4-bromophenyl)-2,6-diphenylpyrimidine (7.54 g, 19.46 mmol) in toluene (100 ml), EtOH (20 ml) and H.sub.2O (20 ml), Pd(PPh.sub.3).sub.4 (1.07 g, 0.93 mmol) and K.sub.3PO.sub.4 (11.80 g, 55.61 mmol) were introduced thereto, and the result was stirred for 5 hours under reflux. After the reaction was completed, the result was cooled to room temperature, and produced solids were filtered and then washed with EA and MeOH. After that, the solids were all dissolved in an excess amount of dichloromethane, and filtered with silica gel to obtain Compound 1 (10 g, 75%).

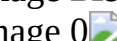
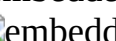
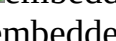
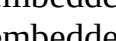
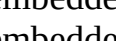
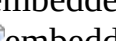
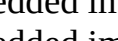
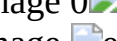
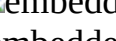
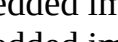
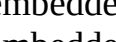
(142) Target compounds were synthesized in the same manner as in Preparation Example 3 using Intermediate C of the following Table 3 instead of 5-bromo-2,3-diphenylthiophene, using Intermediate D instead of 2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)aniline, using

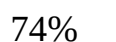
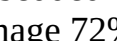
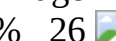
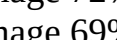
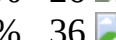
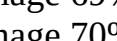
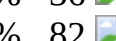
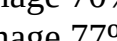
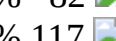
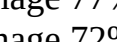
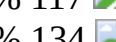
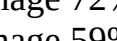
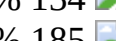
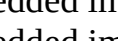
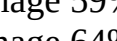
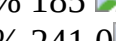
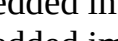
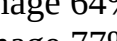
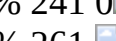
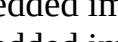
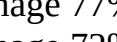
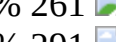
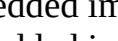
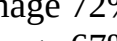
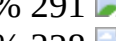
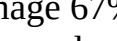
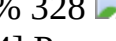
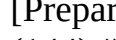
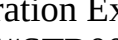
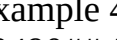
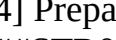
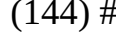
Intermediate E instead of 4-bromobenzoyl chloride, and using Intermediate F instead of 4-(4-bromophenyl)-2,6-diphenylpyrimidine.

(143) TABLE-US-00003 TABLE 3 Intermediate Intermediate Compound C D 6

		9			10
		11			12
		26			29
		31			36
		56			63
		82			104
		108			117
		126			131
		134			149
		181			185
		222			234
		241			242
		248			261
		277			287
		291			294
		301			328

		Intermediate	Intermediate	Compound E	F	6
		9				10
		11				12
		26	01			29
		31	05			36
		56	09			63

		82			104
		108			117
		126			131
		134			149
		181			185
		222			234
		241			242
		248			261
		277			287
		291			294
		301			328

		Compound	Target	Compound	Yield	6
		74%	9			12
		72%	10			11
		81%	11			0
		77%	12			29
		67%	29			31
		64%	31			36
		69%	36			56
		75%	56			63
		77%	63			82
		68%	104			108
		82%	108			0
		77%	117			126
		62%	126			131
		66%	131			149
		70%	149			181
		72%	181			185
		59%	185			222
		62%	222			234
		64%	234			241
		71%	241			0
		73%	242			248
		75%	248			261
		65%	261			277
		72%	277			287
		78%	287			291
		69%	291			294
		67%	294			301
		69%	301			328

[Preparation Example 4] Preparation of Compound 376

(144) ##STR00490## ##STR00491##

Preparation of Compound 376-5

(145) After dissolving 2,5-dibromothiophene (82.82 g, 342.33 mmol) and 2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)aniline (50 g, 228.22 mmol) in 1,4-dioxane (500 ml) and H.sub.2O (100 ml), Pd(PPh.sub.3).sub.4 (7.91 g, 6.85 mmol) and NaHCO.sub.3 (57.52 g, 684.65 mmol) were introduced thereto, and the result was stirred for 8 hours under reflux. After the reaction was completed, the reaction solution was dissolved by introducing MC thereto, and then extracted with distilled water. After drying the organic layer with MgSO.sub.4, the solvent was removed using a rotary evaporator, and then the result was purified by column chromatography using dichloromethane and hexane as a developing solvent to obtain Compound 376-5 (40 g, 69%).

Preparation of Compound 376-4

(146) After dissolving Compound 376-5 (40 g, 157.39 mmol) in MC (400 ml), TEA (47.78 g, 472.16 mmol) was introduced thereto. The temperature was lowered from room temperature to 0° C., and then benzoyl chloride (24.34 g, 173.13 mmol) dissolved in MC was slowly added dropwise thereto. After the reaction was completed, the result was extracted with MC and distilled water, and after drying the organic layer with MgSO.sub.4, the solvent was removed using a rotary evaporator, and then the result was purified by column chromatography using dichloromethane and hexane as a developing solvent to obtain Compound 376-4 (45 g, 80%).

Preparation of Compound 376-3

(147) After dissolving Compound 376-4 (45 g, 125.61 mmol) in nitrobenzene (450 ml), POCl.sub.3 (21.19 g, 138.17 mmol) was slowly added dropwise thereto, and the result was stirred for 4 hours at 150° C. After the reaction was completed, the result was neutralized with an aqueous NaHCO.sub.3 solution and then extracted with MC and distilled water. After drying the organic layer with MgSO.sub.4, the solvent was removed using a rotary evaporator, and then the result was purified by column chromatography using dichloromethane and hexane as a developing solvent to obtain Compound 376-3 (32 g, 75%).

Preparation of Compound 376-2

(148) After dissolving Compound 376-3 (32 g, 94.05 mmol) and phenylboronic acid (12.04 g, 98.75 mmol) in toluene (300 ml), ethanol (60 ml) and H.sub.2O (60 ml), Pd(PPh.sub.3).sub.4 (5.43 g, 4.7 mmol) and K.sub.2CO.sub.3 (39 g, 282.15 mmol) were introduced thereto, and the result was stirred for 8 hours under reflux. After the reaction was completed, the reaction solution was dissolved by introducing MC thereto, and then extracted with distilled water. After drying the organic layer with MgSO.sub.4, the solvent was removed using a rotary evaporator, and then the result was purified by column chromatography using dichloromethane and hexane as a developing solvent to obtain Compound 376-2 (27 g, 85%).

Preparation of Compound 376-1

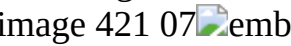
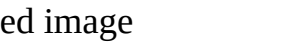
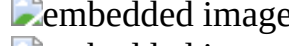
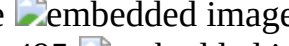
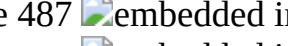
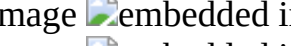
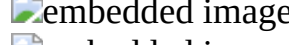
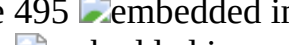
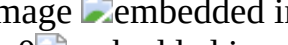
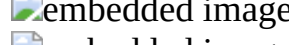
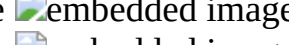
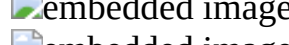
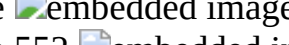
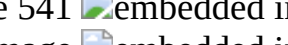
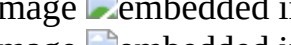
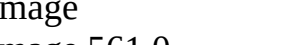
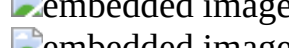
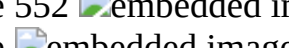
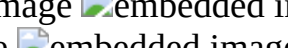
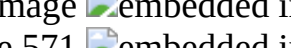
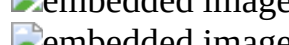
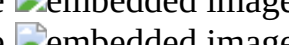
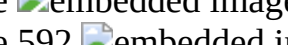
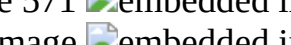
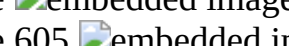
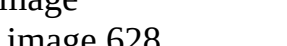
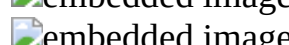
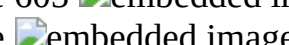
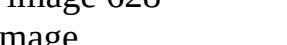
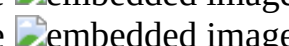
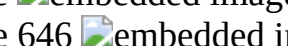
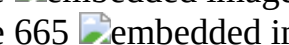
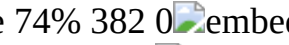
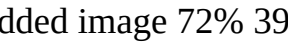
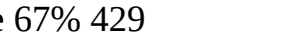
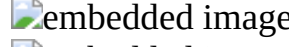
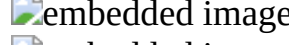
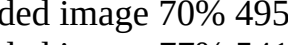
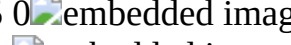
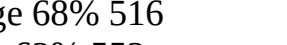
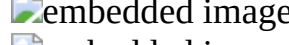
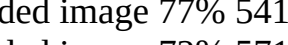
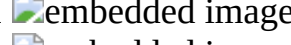
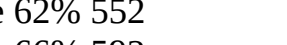
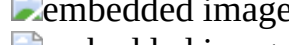
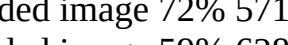
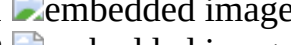
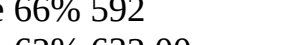
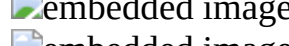
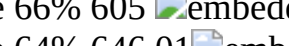
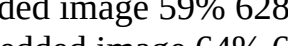
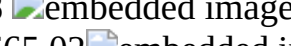
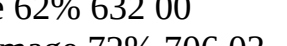
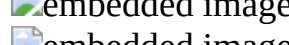
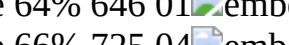
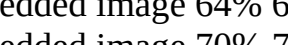
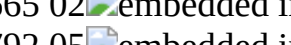
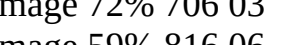
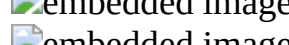
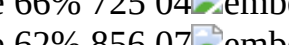
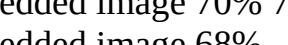
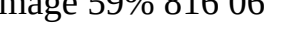
(149) After dissolving Compound 376-2 (27 g, 80.01 mmol) in chloroform (300 ml), the temperature was lowered to 0° C., and bromine (14.07 g, 88.02 mmol) diluted in chloroform was slowly added dropwise thereto. After the reaction was completed, the result was extracted with MC and distilled water. After drying the organic layer with MgSO.sub.4, the solvent was removed using a rotary evaporator, and then the result was purified by column chromatography using dichloromethane and hexane as a developing solvent to obtain Compound 376-1 (29 g, 87%).

Preparation of Compound 376

(150) After dissolving Compound 376-1 (8 g, 19.22 mmol) and 2,4-diphenyl-6-(4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl)-1,3,5-triazine (8.78 g, 20.18 mmol) in toluene (100 ml), ethanol (20 ml) and H.sub.2O (20 ml), Pd(PPh.sub.3).sub.4 (1.11 g, 0.96 mmol) and K.sub.2CO.sub.3 (7.97 g, 57.76 mmol) were introduced thereto, and the result was stirred for 5 hours under reflux. After the reaction was completed, the result was cooled to room temperature, and produced solids were filtered and then washed with EA and MeOH. After that, the solids were all dissolved in an excess amount of dichloromethane, and filtered with silica gel to obtain Compound 376 (9.3 g, 75%).

(151) Target compounds were synthesized in the same manner as in Preparation Example 4 using Intermediate G of the following Table 4 instead of 2,5-dibromothiophene, using Intermediate H instead of phenylboronic acid, and using Intermediate I instead of 2,4-diphenyl-6-(4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl)-1,3,5-triazine.

(152) TABLE-US-00004 TABLE 4 Intermediate Intermediate Intermediate Compound G H I 380

			382		
			390		
			406	01	
			02		
			03		
			419	04	
			05		
			06		
			421	07	
			08		
			09		
			429	0	
			0		
			437		
			463		
			481		
			0		
			487		
			495		
			516		
			526		
			541		
			552		
			561	0	
			571		
			592		
			605		
			0		
			628		
			632		
			646		
			0		
			706		
			725		
			0		
			792	0	
			816		
			856		
			Compound Target Compound Yield	380	
			74%	382	0
			72%	390	
			81%	406	
			77%	419	
			72%	421	
			67%	429	
			64%	437	
			69%	463	
			75%	481	
			77%	487	
			70%	495	0
			68%	516	
			82%	526	
			77%	541	
			62%	552	
			66%	561	
			72%	571	
			66%	592	
			66%	605	
			59%	628	
			62%	632	00
			64%	646	01
			64%	665	02
			72%	706	03
			66%	725	04
			70%	792	05
			59%	816	06
			68%	856	07
			68%		

7.87-8.84 (3H, m), 7.75 (1H, ddd), 7.61-7.40 (15H, m) 56 8.72 (1H, dd), 8.38-8.31 (6H, m), 8.15-8.06 (3H, m), 7.88-7.84 (2H, m) 63 8.42-8.28 (7H, m), 8.24-8.18 (3H, m), 7.94 (1H, s), 7.86-7.74 (9H, m), 7.65 (2H, dd), 7.51-7.32 (15H, m) 82 8.39-8.35 (5H, m), 8.24-8.18 (3H, m), 8.11 (1H, s), 7.94 (1H, dd), 7.85-7.80 (2H, m), 7.72-7.33 (20H, m) 104 9.20-9.12 (4H, m), 8.55 (2H, dd), 8.35 (1H, d), 8.19 (2H, dd), 8.10-8.06 (4H, m), 7.75-7.40 (13H, m), 7.23 (2H, ddd) 108 8.36-8.18 (10H, m), 8.12-8.06 (3H, m), 7.86-7.84 (4H, m), 7.75 (2H, dd), 7.67-7.39 (18H, m) 117 8.55 (2H, dd), 8.25-8.05 (10H, m), 7.66-7.39 (17H, m), 7.22- 7.09 (8H, m) 126 8.77-8.62 (6H, m), 8.39-8.33 (4H, m), 8.25 (2H, d), 8.14 (1H, dd), 7.91-7.83 (7H, m), 7.75 (1H, ddd), 7.60-7.48 (7H, m), 7.25 (2H, d) 131 9.25-9.21 (2H, m), 8.72-8.60 (6H, m), 8.38-8.30 (6H, m), 8.14 (1H, dd), 7.88-7.85 (2H, m), 7.75 (1H, ddd), 7.60-7.37 (7H, m) 134 9.26-9.12 (6H, m), 8.72-8.55 (6H, m), 8.34-8.30 (2H, m), 8.14 (1H, dd), 7.88-7.72 (6H, m), 7.58 (1H, ddd), 7.48-7.39 (2H, m), 7.23 (2H, ddd) 149 9.26-9.12 (6H, m), 8.72-8.53 (4H, m), 8.29-8.35 (4H, m), 8.14 (1H, dd), 7.86 (1H, dd), 7.76-7.69 (4H, m), 7.62-7.57 (2H, m), 7.49-7.39 (2H, m), 7.23 (1H, ddd) 181 8.85 (1H, dd), 8.43-8.28 (6H, m), 8.08-7.93 (6H, m), 7.87-7.73 (5H, m), 7.63-7.31 (16H, m) 185 8.75 (2H, d), 8.64 (2H, d), 8.43-8.33 (5H, m), 8.21-8.11 (3H, m), 7.96-7.72 (10H, m), 7.67-7.48 (7H, m) 222 9.26-9.22 (2H, m), 8.72-8.68 (2H, m), 8.55 (2H, dd), 8.41 (1H, d), 8.35-8.31 (2H, m), 8.21-8.15 (6H, m), 7.94 (1H, dd), 7.75 (1H, dd), 7.67-7.58 (3H, m), 7.54-7.38 (7H, m), 7.22-7.09 (8H, m) 234 9.26-9.22 (2H, m), 8.72-8.68 (2H, m), 8.55 (1H, dd), 8.37-8.17 (13H, m), 7.87-7.78 (3H, m), 7.68-7.38 (13H, m), 7.22-7.08 (4H, m) 241 8.69 (2H, d), 8.37-8.28 (4H, m), 8.23 (1H, s), 8.14 (1H, dd), 7.97-7.72 (10H, m), 7.60-7.31 (15H, m) 242 8.69 (2H, d), 8.39-8.31 (4H, m), 8.25 (2H, d), 8.14 (1H, dd), 7.87-7.72 (6H, m), 7.59-7.31 (15H, m), 7.35-7.23 (3H, m) 248 9.27 (1H, s), 8.79 (1H, dd), 8.69 (2H, d), 8.39-8.28 (4H, m), 8.14 (1H, dd), 7.89-7.30 (18H, m) 261 8.76 (1H, d), 8.39-8.21 (6H, m), 8.06 (1H, d), 7.88-7.80 (3H, m), 7.58-7.30 (17H, m) 277 9.20-9.12 (4H, m), 8.69 (2H, d), 8.55 (2H, dd), 8.16-8.05 (3H, m), 7.88-7.71 (6H, m), 7.62-7.39 (9H, m), 7.23 (2H, ddd) 287 8.39-8.33 (4H, m), 8.27-8.21 (5H, m), 8.11-8.05 (3H, m), 7.62- 7.39 (18H, m), 7.25 (2H, d) 291 8.76 (1H, s), 8.39-8.21 (7H, m), 8.09-8.04 (3H, m), 7.86 (1H, dd), 7.62-36 (17H, m) 294 8.43-8.20 (8H, m), 8.09-8.05 (2H, m), 7.96-7.92 (3H, m), 7.85 (2H, d), 7.75 (1H, dd), 7.62-7.38 (17H, m) 301 8.77-8.61 (6H, m), 8.37-8.28 (4H, m), 8.23 (1H, s), 8.14 (1H, dd), 7.96-7.83 (11H, m), 7.75 (1H, ddd), 7.60-7.47 (7H, m) 328 8.76 (1H, d), 8.60-8.56 (2H, m), 8.39-8.21 (7H, m), 8.07-7.98 (2H, m), 7.88-7.83 (3H, m), 7.58-7.30 (12H, m) 376 8.30-8.28 (6H, m), 8.06 (1H, d), 7.98 (1H, d), 7.85-7.78 (5H, m), 7.60-7.41 (13H, m), 7.25 (2H, d) 380 8.30-8.24 (7H, m), 8.06 (1H, d), 7.98 (1H, d), 7.79-7.78 (3H, m), 7.70 (2H, s), 7.60-7.41 (18H, m) 382 8.30-8.23 (5H, m), 8.06 (1H, d), 7.98 (1H, d), 7.79-7.75 (7H, m), 7.60-7.41 (15H, m) 390 8.30-8.28 (6H, m), 8.16 (1H, d), 8.06 (1H, d), 7.98 (1H, d), 7.84-7.78 (5H, m), 7.60-7.41 (11H, m), 7.25 (2H, d) 406 8.30-8.24 (7H, m), 8.00-7.78 (9H, m), 7.60-7.41 (12H, m), 7.25 (2H, d) 419 8.30-8.24 (3H, m), 8.06-7.74 (11H, m), 7.60-7.39 (8H, m), 7.25 (2H, d) 421 8.30-8.28 (6H, m), 8.06 (1H, d), 7.98 (1H, d), 7.85-7.78 (3H, m), 7.66-7.41 (23H, m), 7.25 (2H, d) 429 8.30-8.23 (7H, m), 8.06 (1H, d), 7.98 (1H, d), 7.85-7.78 (5H, m), 7.60-7.41 (23H, m), 7.25 (4H, d) 437 9.15 (1H, s), 8.93 (2H, d), 8.30-7.98 (13H, m), 7.88-7.78 (5H, m), 7.70 (1H, s), 7.60-7.41 (12H, m) 463 8.45-8.41 (2H, m), 8.30 (2H, d), 8.06 (1H, d), 7.98 (2H, m), 7.83-7.77 (10H, m), 7.60-7.45 (13H, m) 481 8.30-8.28 (6H, m), 8.06 (1H, d), 7.98 (1H, d), 7.85-7.78 (5H, m), 7.60-7.41 (15H, m) 487 8.30-8.23 (5H, m), 8.06 (1H, d), 7.98 (1H, d), 7.79-7.75 (6H, m), 7.60-7.41 (16H, m) 495 8.30-8.28 (6H, m), 8.16 (1H, d), 8.06 (1H, d), 7.98 (1H, d), 7.85-7.78 (5H, m), 7.60-7.41 (13H, m) 516 8.55 (1H, d), 8.42 (1H, d), 8.30-8.23 (7H, m), 8.08-7.98 (4H, m), 7.85-7.78 (5H, m), 7.61-7.41 (13H, m) 526 8.30-8.28 (6H, m), 8.06 (1H, d), 7.98 (1H, d), 7.85-7.78 (5H, m), 7.66-7.41 (23H, m) 541 9.15 (1H, s), 8.93 (2H, d), 8.30-8.28 (6H, m), 8.18-7.98 (6H, m), 7.88-7.78 (9H, m), 7.60-7.41 (10H, m) 552 9.15 (1H, s), 8.93 (2H, d), 8.56 (1H, d), 8.30 (2H, d), 8.18- 7.98 (6H, m), 7.88-7.78 (9H, m), 7.59-7.45 (10H, m), 7.22 (2H, m) 561 8.45-8.41 (2H, m), 8.30-8.23 (8H, m), 8.06 (1H, d), 7.98 (1H, d), 7.85-7.78 (5H, m), 7.60-7.41 (13H, m) 571 8.30-8.28 (6H, m), 8.06 (1H, d), 7.98 (1H, d), 7.89-7.78 (8H, m), 7.60-7.32 (14H, m) 592 8.55 (1H, d), 8.30-8.23 (5H, m), 8.12-8.06 (3H, m), 7.98-7.94 (2H, m), 7.79-7.70 (6H, m), 7.60-7.25 (19H, m) 605 8.30-8.24 (7H, m),

8.06 (1H, d), 7.98 (1H, d), 7.78-7.70 (5H, d), 7.60-7.41 (22H, m) 628 8.30 (2H, d), 8.06-7.92 (5H, m), 7.83-7.78 (10H, m), 7.60-7.45 (13H, m) 632 9.15 (1H, s), 8.93 (2H, d), 8.30-7.98 (13H, m), 7.88-7.70 (7H, m), 7.57-7.41 (11H, m) 646 8.30-8.28 (6H, m), 8.06 (1H, d), 7.98 (1H, d), 7.89-7.78 (8H, m), 7.66-7.34 (14H, m) 665 8.30-8.24 (7H, m), 8.06 (1H, d), 7.98 (1H, d), 7.79-7.78 (3H, m), 7.70 (2H, s), 7.60-7.41 (18H, m) 706 8.30-8.28 (6H, m), 8.10-8.06 (3H, m), 7.98 (1H, d), 7.85-7.78 (3H, m), 7.60-7.41 (13H, m), 7.25 (2H, d) 725 8.30-8.24 (7H, m), 8.06 (1H, d), 798 (1H, d), 7.78 (1H, t), 7.70-7.41 (30H, m) 792 8.30-8.24 (7H, m), 8.10-8.06 (3H, m), 7.98 (1H, d), 7.78 (1H, t), 7.70 (1H, s), 7.60-7.41 (15H, m) 816 9.30 (2H, d), 9.15 (3H, s), 8.93 (2H, d), 8.53 (2H, d), 8.30 (2H, d), 812-7.98 (6H, m), 7.88-7.47 (13H, m), 7.25 (2H, d), 7.14 (2H, t) 856 8.30-8.28 (4H, m), 8.06 (1H, d), 7.98 (1H, d), 7.85-7.78 (7H, m), 7.60-7.41 (15H, m), 7.25 (4H, d)

(155) TABLE-US-00006 TABLE 6 Compound FD-MS Compound FD-MS 1 m/z = 719.91 (C51H33N3S = 719.24) 6 m/z = 720.89 (C50H32N4S = 720.23) 9 m/z = 885.10 (C63H40N4S = 884.30) 10 m/z = 886.09 (C62H39N5S = 885.29) 11 m/z = 667.83 (C47H29N3S = 667.21) 12 m/z = 820.03 (C59H37N3S = 819.27) 26 m/z = 667.83 (C47H29N3S = 667.21) 29 m/z = 644.80 (C44H28N4S = 644.20) 31 m/z = 703.85 (C51H33N3O = 703.26) 36 m/z = 704.83 (C50H32N4O = 704.26) 56 m/z = 651.77 (C47H29N3O = 651.23) 63 m/z = 796.00 (C57H37N3S = 795.27) 82 m/z = 720.89 (C50H32N4S = 720.23) 104 m/z = 628.74 (C44H28N4O = 628.23) 108 m/z = 779.94 (C57H37N3O = 779.29) 117 m/z = 803.97 (C59H37N3O = 803.29) 126 m/z = 722.87 (C48H30N6S = 722.23) 131 m/z = 669.81 (C45H27N5S = 669.20) 134 m/z = 646.77 (C42H26N6S = 646.19) 149 m/z = 646.77 (C42H26N6S = 646.19) 181 m/z = 769.97 (C55H35N3S = 769.26) 185 m/z = 695.84 (C47H29N5S = 695.21) 222 m/z = 805.94 (C57H35N5O = 805.28) 234 m/z = 871.02 (C61H38N6O = 870.31) 241 m/z = 719.91 (C51H33N3S = 719.24) 242 m/z = 720.89 (C50H32N4S = 720.23) 248 m/z = 639.82 (C47H29NS = 639.20) 261 m/z = 667.83 (C47H29N3S = 667.21) 277 m/z = 628.74 (C44H28N4O = 628.23) 287 m/z = 704.83 (C60H32N4O = 704.26) 291 m/z = 651.77 (C47H29N3O = 651.23) 294 m/z = 703.85 (C51H33N3O = 703.26) 301 m/z = 721.88 (C49H31N5S = 721.23) 328 m/z = 653.75 (C45H27N5O = 653.22) 376 m/z = 644.78 (C44H28N4S = 644.20) 380 m/z = 720.88 (C50H32N4S = 720.23) 382 m/z = 643.80 (C45H29N3S = 643.21) 390 m/z = 617.76 (C43H27N3S = 617.19) 406 m/z = 694.84 (C48H30N4S = 694.22) 419 m/z = 580.70 (C40H24N2OS = 580.16) 421 m/z = 796.98 (C56H36N4S = 796.27) 429 m/z = 872.08 (C63H41N3S = 871.30) 437 m/z = 794.96 (C56H34N4S = 794.25) 463 m/z = 719.85 (C47H30NOPS2 = 719.15) 481 m/z = 644.78 (C44H28N4S = 644.20) 487 m/z = 643.80 (C45H29N3S = 643.21) 495 m/z = 617.76 (C43H27N3S = 617.19) 516 m/z = 693.86 (C49H31N3S = 693.22) 526 m/z = 796.98 (C56H36N4S = 796.27) 541 m/z = 794.96 (C56H34N4S = 794.25) 552 m/z = 755.92 (C54H33N3S = 755.24) 561 m/z = 749.94 (C51H31N3S2 = 749.20) 571 m/z = 734.86 592 m/z = 808.99 (C57H36N4S = 808.27) (C50H30N4OS = 734.21) 605 m/z = 796.98 (C56H36N4S = 796.27) 628 m/z = 663.76 (C45H30NOPS = 663.18) 632 m/z = 794.96 (C56H34N4S = 794.25) 646 m/z = 734.86 (C50H30N4OS = 734.21) 665 m/z = 720.88 (C50H32N4S = 720.23) 706 m/z = 628.72 (C44H28N4O = 628.23) 725 m/z = 857.01 (C62H40N4O = 856.32) 792 m/z = 628.72 (C44H28N4O = 628.23) 816 m/z = 778.90 (C56H34N4O = 778.27) 856 m/z = 720.88 (C50H32N4S = 720.23)

<Experimental Example 1> Manufacture of Organic Light Emitting Diode

Comparative Examples 1-1 to 1-3 and Examples 1 to 53

(156) 1) Manufacture of Organic Light Emitting Diode

(157) A transparent ITO electrode thin film obtained from glass for an OLED (manufactured by Samsung-Corning Co., Ltd.) was ultrasonic cleaned using trichloroethylene, acetone, ethanol and distilled water consecutively for 5 minutes each, stored in isopropanol, and used.

(158) Next, an ITO substrate was installed in a substrate folder of a vacuum deposition apparatus, and the following 4,4',4''-tris(N,N-(2-naphthyl)-phenylamino)triphenylamine (2-TNATA) was introduced to a cell in the vacuum deposition apparatus.

(159) ##STR00608##

(160) Subsequently, the chamber was evacuated until the degree of vacuum therein reached 10.sup.-6 torr, and then 2-TNATA was evaporated by applying a current to the cell to deposit a hole injection layer having a thickness of 600 Å on the ITO substrate.

(161) To another cell in the vacuum deposition apparatus, the following N,N'-bis(α-naphthyl)-N,N'-diphenyl-4,4'-diamine (NPB) was introduced, and evaporated by applying a current to the cell to deposit a hole transfer layer having a thickness of 300 Å on the hole injection layer.

(162) ##STR00609##

(163) After forming the hole injection layer and the hole transfer layer as above, a blue light emitting material having a structure as below was deposited thereon as a light emitting layer. Specifically, in one side cell in the vacuum deposition apparatus, H1, a blue light emitting host material, was vacuum deposited to a thickness of 200 Å, and D1, a blue light emitting dopant material, was vacuum deposited thereon by 5% with respect to the host material.

(164) ##STR00610##

(165) Subsequently, as an electron transfer layer, a compound shown in Table 6 was deposited to a thickness of 300 Å.

(166) ##STR00611##

(167) As an electron injection layer, lithium fluoride (LiF) was deposited to a thickness of 10 Å, and an Al cathode was deposited to a thickness of 1,000 Å, and as a result, an OLED was manufactured.

(168) Meanwhile, all the organic compounds required to manufacture the OLED were vacuum sublimation purified under 10.sup.-6 torr to 10.sup.-8 torr by each material to be used in the OLED manufacture.

(169) 2) Driving Voltage and Light Emission Efficiency of Organic Light Emitting Diode

(170) For the organic light emitting diodes manufactured as above, electroluminescent (EL) properties were measured using M7000 manufactured by McScience Inc., and with the measurement results, 195 was measured when standard luminance was 700 cd/m.sup.2 using a lifetime measurement system (M6000) manufactured by McScience Inc. Results of measuring driving voltage, light emission efficiency, external quantum efficiency and color coordinate (CIE) of the blue organic light emitting diode are as shown in Table 7.

(171) TABLE-US-00007 TABLE 7 Light Driving Emission Com- Voltage Efficiency Lifetime
pound (V) (cd/A) CIE (x, y) (T95) Comparative E1 5.56 5.91 (0.134, 0.100) 28 Example 1-1
Comparative BBQB 5.50 6.10 (0.134, 0.101) 30 Example 1-2 Comparative TBQB 5.51 6.15
(0.134, 0.102) 29 Example 1-3 Example 1 1 5.11 6.95 (0.134, 0.101) 32 Example 2 6 4.96 6.84
(0.134, 0.102) 47 Example 3 9 5.14 6.79 (0.134, 0.101) 33 Example 4 10 4.99 6.88 (0.134, 0.103)
49 Example 5 29 5.45 6.12 (0.134, 0.102) 34 Example 6 31 5.03 7.09 (0.134, 0.101) 30 Example 7
36 4.72 7.12 (0.134, 0.102) 52 Example 8 63 5.32 6.33 (0.134, 0.101) 29 Example 9 82 5.40 6.13
(0.134, 0.101) 31 Example 10 104 5.30 6.72 (0.134, 0.100) 33 Example 11 108 5.37 6.35 (0.134,
0.101) 32 Example 12 126 5.38 6.41 (0.134, 0.100) 30 Example 13 134 5.27 6.42 (0.134, 0.100) 31
Example 14 149 5.48 6.21 (0.134, 0.100) 32 Example 15 181 4.72 7.39 (0.134, 0.100) 49 Example
16 185 5.45 6.68 (0.134, 0.100) 34 Example 17 234 5.22 6.28 (0.134, 0.102) 37 Example 18 241
5.12 6.20 (0.134, 0.101) 32 Example 19 242 5.09 6.97 (0.134, 0.102) 56 Example 20 277 5.42 6.88
(0.134, 0.100) 25 Example 21 287 5.21 5.45 (0.134, 0.103) 37 Example 22 294 5.38 6.66 (0.134,
0.100) 35 Example 23 301 5.40 6.36 (0.134, 0.102) 32 Example 24 376 4.72 7.11 (0.134, 0.102) 45
Example 25 380 5.19 6.13 (0.134, 0.100) 33 Example 26 382 5.22 6.21 (0.134, 0.102) 32 Example
27 390 5.21 6.21 (0.134, 0.101) 35 Example 28 406 5.31 6.13 (0.134, 0.102) 36 Example 29 419
5.21 6.19 (0.134, 0.100) 33 Example 30 421 4.67 6.49 (0.134, 0.102) 41 Example 31 429 5.11 6.03
(0.134, 0.101) 34 Example 32 437 5.09 6.32 (0.134, 0.102) 35 Example 33 463 5.77 5.73 (0.134,
0.102) 58 Example 34 481 4.62 6.44 (0.134, 0.101) 44 Example 35 487 4.55 7.30 (0.134, 0.102) 21
Example 36 495 5.22 6.11 (0.134, 0.101) 36 Example 37 516 5.65 5.69 (0.134, 0.103) 61 Example
38 526 5.12 5.99 (0.134, 0.102) 35 Example 39 541 4.83 6.87 (0.134, 0.102) 46 Example 40 552

5.20 6.17 (0.134, 0.101) 33 Example 41 561 5.32 6.16 (0.134, 0.102) 36 Example 42 571 5.15 5.89 (0.134, 0.101) 35 Example 43 592 5.26 6.08 (0.134, 0.102) 37 Example 44 605 5.21 6.10 (0.134, 0.101) 36 Example 45 628 5.74 5.79 (0.134, 0.102) 60 Example 46 632 5.23 6.12 (0.134, 0.101) 31 Example 47 646 5.32 6.23 (0.134, 0.102) 36 Example 48 665 5.12 6.53 (0.134, 0.101) 35 Example 49 706 5.18 6.05 (0.134, 0.102) 33 Example 50 725 5.11 5.96 (0.134, 0.101) 34 Example 51 792 5.27 5.99 (0.134, 0.103) 37 Example 52 816 5.23 6.03 (0.134, 0.101) 36 Example 53 856 4.81 7.13 (0.134, 0.102) 47

(172) As seen from the results of Table 7, the organic light emitting diode using the electron transfer layer material of the blue organic light emitting diode of the present disclosure had lower driving voltage and significantly improved light emission efficiency and lifetime compared to Comparative Example 1. Particularly, it was identified that Compounds 6, 10, 36, 181, 242, 376, 421, 481, 541 and 856 were superior in all aspects of driving, efficiency and lifetime.

(173) Such a result is considered to be due to the fact that, when using the disclosed compound having proper length and strength, and flatness as an electron transfer layer, a compound in an excited state is made by receiving electrons under a specific condition, and particularly when a hetero-skeleton site of the compound is formed in an excited state, excited energy moves to a stable state before the excited hetero-skeleton site goes through other reactions, and a relatively stabilized compound is capable of efficiently transferring electrons without the compound being decomposed or destroyed. For reference, those that are stable when excited are considered to be aryl or acene-based compounds or polycyclic hetero-compounds. Accordingly, it is considered that excellent results in all aspects of driving, efficiency and lifetime were obtained by the compound of the present disclosure enhancing enhanced electron-transfer properties or improved stability.

<Experimental Example 2> Manufacture of Organic Light Emitting Diode

Comparative Example 2

(174) 1) Manufacture of Organic Light Emitting Diode

(175) A transparent ITO electrode thin film obtained from glass for an OLED (manufactured by Samsung-Corning Co., Ltd.) was ultrasonic cleaned using trichloroethylene, acetone, ethanol and distilled water consecutively for 5 minutes each, stored in isopropanol, and used.

(176) Next, an ITO substrate was installed in a substrate folder of a vacuum deposition apparatus, and the following 4,4',4''-tris(N,N-(2-naphthyl)-phenylamino)triphenylamine (2-TNATA) was deposited to a cell in the vacuum deposition apparatus.

(177) ##STR00612##

(178) Subsequently, the chamber was evacuated until the degree of vacuum therein reached 10.sup.-6 torr, and then 2-TNATA was evaporated by applying a current to the cell to deposit a hole injection layer having a thickness of 600 Å on the ITO substrate.

(179) To another cell in the vacuum deposition apparatus, the following N,N'-bis(α-naphthyl)-N,N'-diphenyl-4,4'-diamine (NPB) was introduced, and evaporated by applying a current to the cell to deposit a hole transfer layer having a thickness of 300 Å on the hole injection layer.

(180) ##STR00613##

(181) After forming the hole injection layer and the hole transfer layer as above, a blue light emitting material having a structure as below was deposited thereon as a light emitting layer. Specifically, in one side cell in the vacuum deposition apparatus, H1, a blue light emitting host material, was vacuum deposited to a thickness of 200 Å, and D1, a blue light emitting dopant material, was vacuum deposited thereon by 5% with respect to the host material.

(182) ##STR00614##

(183) Then, as an electron transfer layer, a compound having the following structural formula E1 was deposited to a thickness of 300 Å.

(184) ##STR00615##

(185) As an electron injection layer, lithium fluoride (LiF) was deposited to a thickness of 10 Å, and an Al cathode was deposited to a thickness of 1,000 Å, and as a result, an OLED was

manufactured.

(186) Meanwhile, all the organic compounds required to manufacture the OLED were vacuum sublimation purified under 10.sup.-6 torr to 10.sup.-8 torr by each material to be used in the OLED manufacture.

Examples 54 to 63

(187) Organic electroluminescent diodes were manufactured in the same manner as in Comparative Example 2 except that E1 was formed to a thickness of 250 Å as the electron transfer layer, and on the electron transfer layer, a hole blocking layer having a thickness of 50 Å was formed using a compound presented in Table 8.

(188) Results of measuring driving voltage, light emission efficiency, color coordinate (CIE) and lifetime of the blue organic light emitting diode manufactured according to the present disclosure are as shown in Table 8.

(189) TABLE-US-00008 TABLE 8 Light Driving Emission Com- Voltage Efficiency Lifetime
pound (V) (cd/A) CIE (x, y) (T95) Comparative — 5.50 5.57 (0.134, 0.100) 31 Example 2
Example 54 12 4.87 6.17 (0.134, 0.101) 51 Example 55 117 4.73 6.59 (0.134, 0.102) 56 Example
56 222 5.11 6.65 (0.134, 0.101) 55 Example 57 248 4.87 6.54 (0.134, 0.103) 61 Example 58 406
4.82 6.44 (0.134, 0.103) 54 Example 59 421 4.95 6.59 (0.134, 0.103) 56 Example 60 437 5.11 6.45
(0.134, 0.103) 57 Example 61 463 4.85 6.31 (0.134, 0.103) 55 Example 62 541 4.92 6.67 (0.134,
0.103) 58 Example 63 571 4.87 6.55 (0.134, 0.103) 49

(190) As seen from the results of Table 8, the organic electroluminescent diode using the hole blocking layer material of the blue organic electroluminescent diode of the present disclosure had lower driving voltage and significantly improved light emission efficiency and lifetime compared to Comparative Example 2.

<Experimental Example 3> Manufacture of Organic Light Emitting Diode

Comparative Example 3

(191) 1) Manufacture of Organic Light Emitting Diode

(192) A transparent ITO electrode thin film obtained from glass for an OLED (manufactured by Samsung-Corning Co., Ltd.) was ultrasonic cleaned using trichloroethylene, acetone, ethanol and distilled water consecutively for 5 minutes each, stored in isopropanol, and used.

(193) Next, an ITO substrate was installed in a substrate folder of a vacuum deposition apparatus, and the following 4,4',4''-tris(N,N-(2-naphthyl)-phenylamino)triphenylamine (2-TNATA) was introduced to a cell in the vacuum deposition apparatus.

(194) ##STR00616##

(195) Subsequently, the chamber was evacuated until the degree of vacuum therein reached 10.sup.-6 torr, and then 2-TNATA was evaporated by applying a current to the cell to deposit a hole injection layer having a thickness of 600 Å on the ITO substrate.

(196) To another cell in the vacuum deposition apparatus, the following N,N'-bis(α-naphthyl)-N,N'-diphenyl-4,4'-diamine (NPB) was introduced, and evaporated by applying a current to the cell to deposit a hole transfer layer having a thickness of 300 Å on the hole injection layer.

(197) ##STR00617##

(198) The hole injection layer and the hole transfer layer were formed as above.

(199) As a light emitting layer, H2, a host, and Ir(ppy).sub.3, a green phosphorescent dopant, by 7% doping were deposited to 400 Å thereon.

(200) ##STR00618##

(201) After that, BCP was deposited to 60 Å as a hole blocking layer, and Alq.sub.3 was deposited to 200 Å thereon as an electron transfer layer. Lastly, an electron injection layer was formed on the electron transfer layer by depositing lithium fluoride (LiF) to a thickness of 10 Å, and then a cathode was formed on the electron injection layer by depositing an aluminum (Al) cathode to a thickness of 1,200 Å, and as a result, an organic electroluminescent diode was manufactured.

(202) Meanwhile, all the organic compounds required to manufacture the OLED were vacuum

sublimation purified under 10.sup.-6 torr to 10.sup.-8 torr for each material to be used in the OLED manufacture.

Examples 64 to 69

(203) Organic electroluminescent diodes were manufactured in the same manner as in Comparative Example 3 except that, when forming the host, a compound presented in Table 9 was used instead of H2.

(204) Results of measuring driving voltage, light emission efficiency, color coordinate (CIE) and lifetime of the green organic light emitting diode manufactured according to the present disclosure are as shown in Table 9.

(205) TABLE-US-00009 TABLE 9 Light Driving Emission Com- Voltage Efficiency Lifetime
pound (V) (cd/A) CIE (x, y) (T95) Comparative H2 5.24 35.83 (0.266, 0.711) 63 Example 3
Example 64 421 5.11 41.23 (0.266, 0.712) 92 Example 65 437 4.96 43.88 (0.265, 0.711) 79
Example 66 571 4.74 46.02 (0.265, 0.712) 89 Example 67 592 5.12 38.82 (0.266, 0.713) 92
Example 68 632 4.89 39.44 (0.266, 0.711) 93 Example 69 646 4.72 41.36 (0.267, 0.711) 77
<Experimental Example 4> Manufacture of Organic Light Emitting Diode

Comparative Example 4 and Examples 70 to 79

(206) 1) Manufacture of Organic Light Emitting Diode

(207) A transparent ITO electrode thin film obtained from glass for an OLED (manufactured by Samsung-Corning Co., Ltd.) was ultrasonic cleaned using trichloroethylene, acetone, ethanol and distilled water consecutively for 5 minutes each, stored in isopropanol, and used.

(208) On the transparent ITO electrode (anode), an organic material was formed in a 2-stack white organic light emitting diode (WOLED) structure. As for the first stack, TAPC was thermal vacuum deposited to a thickness of 300 Å first to form a hole transfer layer. After forming the hole transfer layer, a light emitting layer was thermal vacuum deposited thereon as follows. The light emitting layer was deposited to 300 Å by doping Flrpic to TCz1, a host, by 8% as a blue phosphorescent dopant. After forming an electron transfer layer to 400 Å using TmPyPB, a charge generation layer was formed to 100 Å by doping Cs.sub.2CO.sub.3 to a compound described in the following Table 10 by 20%.

(209) As for the second stack, MoO.sub.3 was thermal vacuum deposited to a thickness of 50 Å first to form a hole injection layer. A hole transfer layer, a common layer, was formed by doping MoO.sub.3 to TAPC by 20% to 100 Å and then depositing TAPC to 300 Å. A light emitting layer was deposited thereon to 300 Å by doping Ir(ppy).sub.3, a green phosphorescent dopant, to TCz1, a host, by 8%, and an electron transfer layer was formed to 600 Å using TmPyPB. Lastly, an electron injection layer was formed on the electron transfer layer by depositing lithium fluoride (LiF) to a thickness of 10 Å, and then a cathode was formed on the electron injection layer by depositing an aluminum (Al) cathode to a thickness of 1,200 Å, and as a result, an organic light emitting diode was manufactured.

(210) Meanwhile, all the organic compounds required to manufacture the OLED were vacuum sublimation purified under 10.sup.-6 torr to 10.sup.-8 torr by each material to be used in the OLED manufacture.

(211) ##STR00619## ##STR00620##

(212) Results of measuring driving voltage, light emission efficiency, color coordinate (CIE) and lifetime of the white organic light emitting diodes manufactured according to the present disclosure are as shown in Table 10.

(213) TABLE-US-00010 TABLE 10 Light Driving Emission Com- Voltage Efficiency Lifetime
pound (V) (cd/A) CIE (x, y) (T95) Comparative BPhen 7.54 54.23 (0.213, 0.430) 25 Example 4
Example 70 11 6.84 60.44 (0.212, 0.421) 35 Example 71 26 6.44 66.32 (0.211, 0.433) 40 Example
72 29 6.92 59.68 (0.214, 0.439) 31 Example 73 56 6.27 67.99 (0.212, 0.424) 53 Example 74 104
6.82 59.18 (0.214, 0.437) 34 Example 75 131 6.73 62.44 (0.212, 0.426) 37 Example 76 134 6.82
57.18 (0.214, 0.437) 34 Example 77 261 6.29 68.49 (0.213, 0.424) 51 Example 78 291 6.37 67.41

(0.213, 0.423) 46 Example 79 328 6.41 66.31 (0.212, 0.421) 45

(214) As seen from the results of Table 10, the organic electroluminescent diode using the charge generation layer material of the 2-stack white organic electroluminescent diode of the present disclosure had lower driving voltage and improved light emission efficiency compared to Comparative Example 3. Such a result is considered to be due to the fact that the compound of the present disclosure used as an N-type charge generation layer formed with the disclosed skeleton having proper length and strength, and flatness and a proper hetero-compound capable of binding to metals forms a gap state in the N-type charge generation layer by doping an alkali metal or an alkaline earth metal, and electrons produced from a P-type charge generation layer are readily injected into an electron transfer layer through the gap state produced in the N-type charge generation layer. Accordingly, the P-type charge generation layer may favorably inject and transfer electrons to the N-type charge generation layer, and as a result, driving voltage was lowered, and efficiency and lifetime were improved in the organic light emitting diode.

(215) Hereinbefore, preferred examples of the present disclosure have been described in detail, however, the scope of a right of the present disclosure is not limited thereto, and various modifications and improvements made by those skilled in the art using the basic concept of the present disclosure defined in the attached claims also fall within the scope of a right of the present disclosure.

Claims

1. A compound represented by one of the following Chemical Formulae 2 to 4: ##STR00621## wherein, in Chemical Formulae 2 to 4, X^{sup.1} is —O—; or —S—; Ar^{sup.1} to Ar^{sup.5} are each independently hydrogen; deuterium; a cyano group; —P(=O) RR'; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group; in chemical formulae 2 to 4, L^{sup.1} to L^{sup.3} are each independently a single bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group; in chemical formulae 2 to 4, L^{sup.4} and L^{sup.5} are each independently a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group; R and R' are each independently hydrogen; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group; n1 to n5 are each independently one of integers of 1 to 4; and at least one of Ar^{sup.1} to Ar^{sup.5} is a substituent having electron properties; or a substituent having hole properties, wherein the substituent having hole properties is a substituted or unsubstituted C6 to C60 aryl group having hole properties, a substituted or unsubstituted C2 to C60 heteroaryl group having hole properties, a substituted or unsubstituted arylamine group, or a substituted or unsubstituted heteroarylamine group, wherein the substituent having electron properties is a substituted or unsubstituted C2 to C60 heteroaryl group having electron properties; and the substituted or unsubstituted C2 to C60 heteroaryl group having electron properties is a substituted or unsubstituted imidazolyl group, a substituted or unsubstituted tetrazolyl group, a substituted or unsubstituted quinolinylene group, a substituted or unsubstituted isoquinolinylene group, a substituted or unsubstituted pyridinylene group, a substituted or unsubstituted pyrimidinylene group, a substituted or unsubstituted triazinylene group, a substituted or unsubstituted furanyl group, a substituted or unsubstituted benzofuranyl group, a substituted or unsubstituted isofuranyl group, a substituted or unsubstituted benzoisofuranyl group, a substituted or unsubstituted oxazoline group, a substituted or unsubstituted benzoxazoline group, a substituted or unsubstituted oxadiazoline group, a substituted or unsubstituted benzoxadiazoline group, a substituted or unsubstituted oxatriazolyl group, a substituted or unsubstituted thiophenyl group, a substituted or unsubstituted benzothiophenyl group, a substituted or unsubstituted isothiazoline group, a substituted or unsubstituted

benzothiazoline group, a substituted or unsubstituted thiazoline group, a substituted or unsubstituted benzothiazoline group, a substituted or unsubstituted pyridazinyl group, a substituted or unsubstituted benzopyridazinyl group, a substituted or unsubstituted pyrazinyl group, a substituted or unsubstituted benzopyrazinyl group, a substituted or unsubstituted phthalazinyl group, a substituted or unsubstituted benzoquinolinyl group, a substituted or unsubstituted quinoxalinyl group, a substituted or unsubstituted quinazolinyl group, a substituted or unsubstituted acridinyl group, a substituted or unsubstituted phenanthrolinyl group, a substituted or unsubstituted phenazinyl group, or a combination thereof, wherein “substituted or unsubstituted” means being substituted with one or more substituents selected from the group consisting of deuterium, —CN, —P(=O) RR', a C1 to C20 linear or branched alkyl group, a C6 to C60 monocyclic or polycyclic aryl group and a C2 to C60 monocyclic or polycyclic heteroaryl group, or being unsubstituted.

2. The compound of claim 1, wherein the compound represented by one of Chemical Formulae 2 to 4 is one of compounds of the following Group I: ##STR00622## ##STR00623## ##STR00624##

##STR00625## ##STR00626## ##STR00627## ##STR00628## ##STR00629## ##STR00630## ##STR00631## ##STR00632## ##STR00633## ##STR00634## ##STR00635## ##STR00636## ##STR00637## ##STR00638## ##STR00639## ##STR00640## ##STR00641## ##STR00642## ##STR00643## ##STR00644## ##STR00645## ##STR00646## ##STR00647## ##STR00648## ##STR00649## ##STR00650## ##STR00651## ##STR00652## ##STR00653## ##STR00654## ##STR00655## ##STR00656## ##STR00657## ##STR00658## ##STR00659## ##STR00660## ##STR00661## ##STR00662## ##STR00663## ##STR00664## ##STR00665## ##STR00666## ##STR00667## ##STR00668## ##STR00669## ##STR00670## ##STR00671## ##STR00672## ##STR00673## ##STR00674## ##STR00675## ##STR00676## ##STR00677## ##STR00678## ##STR00679## ##STR00680## ##STR00681## ##STR00682## ##STR00683## ##STR00684## ##STR00685## ##STR00686## ##STR00687## ##STR00688## ##STR00689## ##STR00690## ##STR00691## ##STR00692## ##STR00693## ##STR00694## ##STR00695## ##STR00696## ##STR00697## ##STR00698## ##STR00699## ##STR00700## ##STR00701## ##STR00702## ##STR00703## ##STR00704## ##STR00705## ##STR00706## ##STR00707## ##STR00708## ##STR00709## ##STR00710## ##STR00711## ##STR00712## ##STR00713## ##STR00714## ##STR00715## ##STR00716## ##STR00717## ##STR00718## ##STR00719## ##STR00720## ##STR00721## ##STR00722## ##STR00723## ##STR00724## ##STR00725## ##STR00726## ##STR00727## ##STR00728## ##STR00729## ##STR00730## ##STR00731## ##STR00732## ##STR00733## ##STR00734## ##STR00735## ##STR00736## ##STR00737## ##STR00738## ##STR00739## ##STR00740## ##STR00741## ##STR00742## ##STR00743## ##STR00744## ##STR00745## ##STR00746## ##STR00747## ##STR00748## ##STR00749## ##STR00750## ##STR00751## ##STR00752## ##STR00753## ##STR00754## ##STR00755## ##STR00756## ##STR00757## ##STR00758## ##STR00759## ##STR00760## ##STR00761## ##STR00762## ##STR00763## ##STR00764## ##STR00765## ##STR00766## ##STR00767## ##STR00768## ##STR00769## ##STR00770## ##STR00771## ##STR00772## ##STR00773## ##STR00774## ##STR00775## ##STR00776## ##STR00777## ##STR00778## ##STR00779## ##STR00780## ##STR00781## ##STR00782## ##STR00783## ##STR00784## ##STR00785## ##STR00786## ##STR00787## ##STR00788## ##STR00789## ##STR00790## ##STR00791## ##STR00792## ##STR00793## ##STR00794## ##STR00795## ##STR00796## ##STR00797## ##STR00798## ##STR00799## ##STR00800## ##STR00801## ##STR00802## ##STR00803## ##STR00804## ##STR00805## ##STR00806## ##STR00807## ##STR00808## ##STR00809## ##STR00810## ##STR00811## ##STR00812## ##STR00813## ##STR00814## ##STR00815## ##STR00816## ##STR00817## ##STR00818## ##STR00819## ##STR00820## ##STR00821## ##STR00822## ##STR00823## ##STR00824## ##STR00825## ##STR00826## ##STR00827## ##STR00828## ##STR00829## ##STR00830## ##STR00831## ##STR00832## ##STR00833## ##STR00834## ##STR00835## ##STR00836## ##STR00837## ##STR00838## ##STR00839## ##STR00840## ##STR00841## ##STR00842## ##STR00843## ##STR00844## ##STR00845## ##STR00846##

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3. An organic optoelectronic diode comprising: an anode and a cathode facing each other; and at least one organic layer disposed between the anode and the cathode, wherein the organic layer comprises the compound of claim 1.

4. The organic optoelectronic diode of claim 3, wherein the organic layer comprises an electron transfer layer, a charge generation layer, a hole blocking layer or a combination thereof, and the electron transfer layer, the charge generation layer or the hole blocking layer comprises the compound.

5. A display device comprising the organic optoelectronic diode of claim 3.
